

FC7xxx OCU User Manual

Rev.0.6

Revision History

Revision	Date	Changes
0.1	2023/10/20	Initial release for MCAL V0.3.0
0.6	2023/03/29	Updated for MCAL V0.6.0 - Added support for FC7240

Table of Contents

Revision History	2
Table of Contents	3
Chapter 1 OCU Introduction	5
1.1 Requirement Tracing	5
1.2 Hardware Summary	5
Chapter 2 Software Design	6
2.1 Rejected Requirements	6
2.2 File Structure	7
2.3 Macros	8
2.3.1 Macros in Ocu.h	8
2.3.2 Macros in Ocu_Ftu_Types.h	10
2.3.3 Macros in Ocu_Version.h	11
2.3.4 Macros in Ocu_Cfg.h	11
2.4 Enums	14
2.4.1 Enumerations in Ocu.h	14
2.4.2 Enumerations in Ocu_Types.h	14
2.4.3 Enumerations in Ocu_Ftu_Types.h	15
2.5 Typedefs	16
2.5.1 Typedefs in Ocu_Types.h	16
2.5.2 Typedefs in Ocu_Cfg.h	16
2.5.3 Typedefs in Ocu_Ftu_Types.h	16
2.6 Structures	16
2.6.1 Ocu_ConfigType	16
2.6.2 Ocu_ChannelConfigType	17
2.6.3 Ocu_HwConfigType	17
2.6.4 Ocu_HwChannelConfigType	17
2.6.5 Ocu_FtuConfigType	18
2.6.6 Ocu_Ftu_ChannelConfigType	18
2.6.7 Ocu_Ftu_ModuleConfigType	18
2.6.8 Ocu_SetThresholdType	19
2.7 API Functions	19
2.7.1 Functions in Ocu.h	19
2.8 Hardware Functions	24
2.8.1 Functions in Ocu_Hw.h	24
2.8.2 Functions in Ocu_Ftu.h	26

2.9	API Sequence Diagram	28
2.9.1	Initialization	28
2.9.2	De-initialization.....	29
2.9.3	Using the Ocu Notifications.....	29
2.9.4	Ocu_SetPinState.....	30
2.9.5	Ocu_SetPinAction	31
Chapter 3 Tresos Configuration Items		32
3.1	Container Inclusion Relation.....	32
3.1.1	IMPLEMENTATION_CONFIG_VARIANT	32
3.1.2	OcuConfigurationOfOptionalApis.....	32
3.1.3	OcuConfigSet	33
3.1.4	OcuChannel.....	33
3.1.5	OcuHWSpecificSettings.....	33
3.1.6	OcuGeneral.....	34
3.1.7	CommonPublishedInformation.....	34
3.2	Containers and Variables	34
3.2.1	IMPLEMENTATION_CONFIG_VARIANT	34
3.2.2	OcuConfigurationOfOptionalApis.....	34
3.2.3	OcuConfigSet	37
3.2.4	OcuChannel	37
3.2.5	OcuHWSpecificSettings.....	41
3.2.6	OcuGeneral.....	42
3.2.7	CommonPublishedInformation.....	44
Chapter 4 Configuration Guides.....		47
4.1	Configuration Item Constraint.....	47
4.2	OCU Usage Common Steps.....	50

Chapter 1 OCU Introduction

1.1 Requirement Tracing

The design of this module follows the specifications of the OCU driver specified in AUTOSAR Classic Platform Release 4.6.0. For detailed requirements, refer to the *AUTOSAR_SWS_OCUDriver*.

1.2 Hardware Summary

The OCU driver provides logical channels which are configured by Flexible Timer Unit (FTU) hardware.

The OCU driver provides several services including initializing channels, de-initializing channels, starting the channels, stopping the channels, and others.

The OCU driver implements up to 96 channels on the FC7xxx FTU peripheral.

Chapter 2 Software Design

2.1 Rejected Requirements

Rejected Requirement 3 **SWS_Ocu_00020**

Description	The detection of production errors cannot be switched off.
Rejection Reason	Current implementation of Ocu driver DEM errors are not used

Rejected Requirement 4 **SWS_Ocu_00023**

Description	Each variable that shall be accessible by AUTOSAR Debugging, shall be defined as global variable.
Rejection Reason	AUTOSAR Debugging concept is not implemented.

Rejected Requirement 4 **SWS_Ocu_00024**

Description	All type definitions of variables which shall be debugged shall be accessible by the header file Ocu.h.
Rejection Reason	AUTOSAR Debugging concept is not implemented.

Rejected Requirement 5 **SWS_Ocu_00025**

Description	The declaration of variables in the header file shall be such that it is possible to calculate the size of the variables by C-"sizeof".
Rejection Reason	AUTOSAR Debugging concept is not implemented.

Rejected Requirement 6 **SWS_Ocu_00026**

Description	Variables available for debugging shall be described in the respective OCU driver Description.
Rejection Reason	AUTOSAR Debugging concept is not implemented. In par with PRMCAL-3330

Rejected Requirement 6 **SWS_Ocu_00125**

Description	If source code for caller and callee of Ocu_GetVersionInfo is available; the OCU driver should realize Ocu_GetVersionInfo as a macro, defined in the module' s header file.
Rejection Reason	Ocu_GetVersionInfo is defined as a function as a common approach in OCU MCAL driver.

Rejected Requirement 6 **SWS_Ocu_00155**

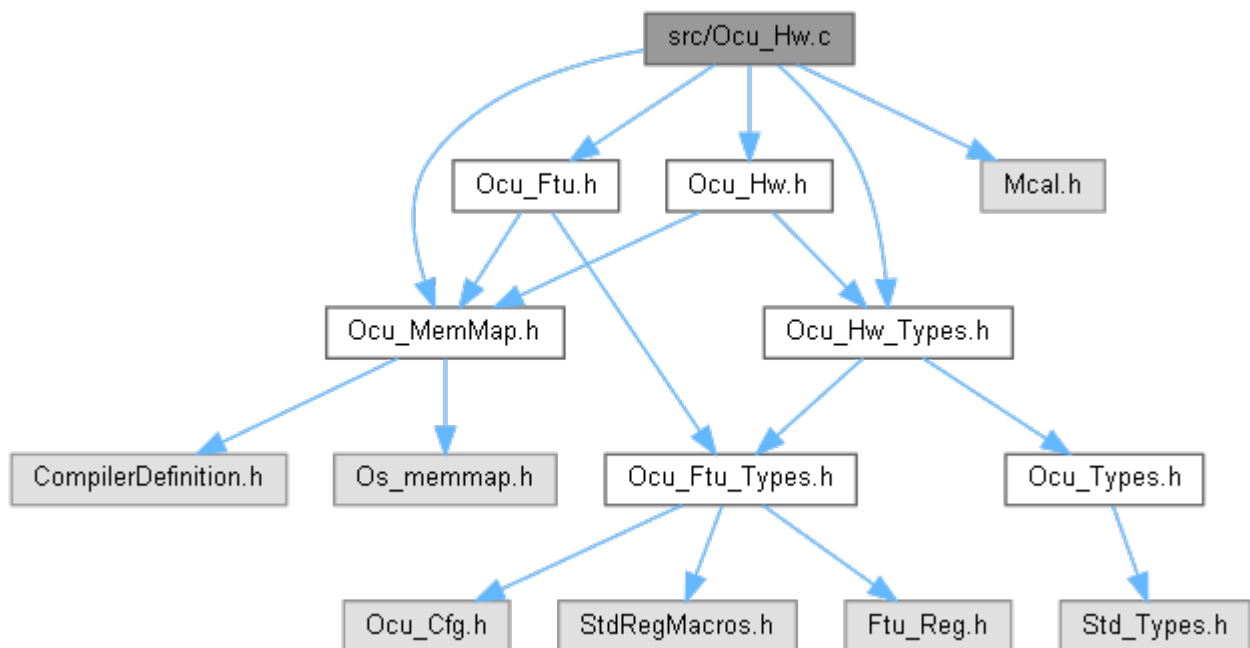
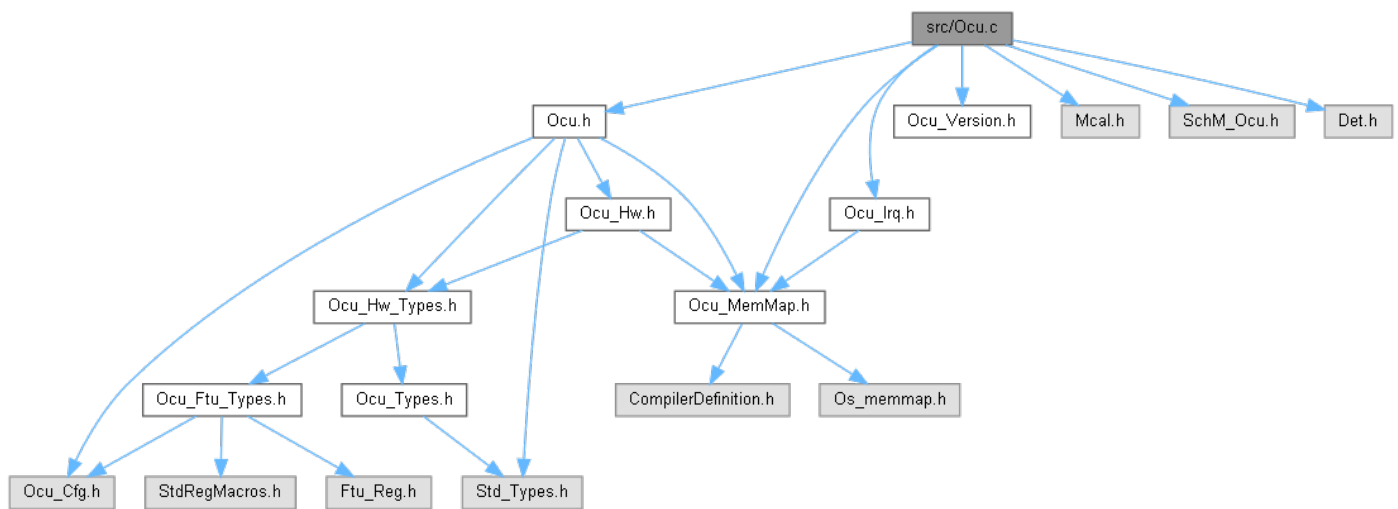
Description	This parameter is used to allow the OCU channel to trigger an ADC channel upon compare match, if this is supported by hardware. The value of the parameter represents the ADC physical channel to trigger.
Rejection Reason	Trigger for ADC has not supported in OCU driver.

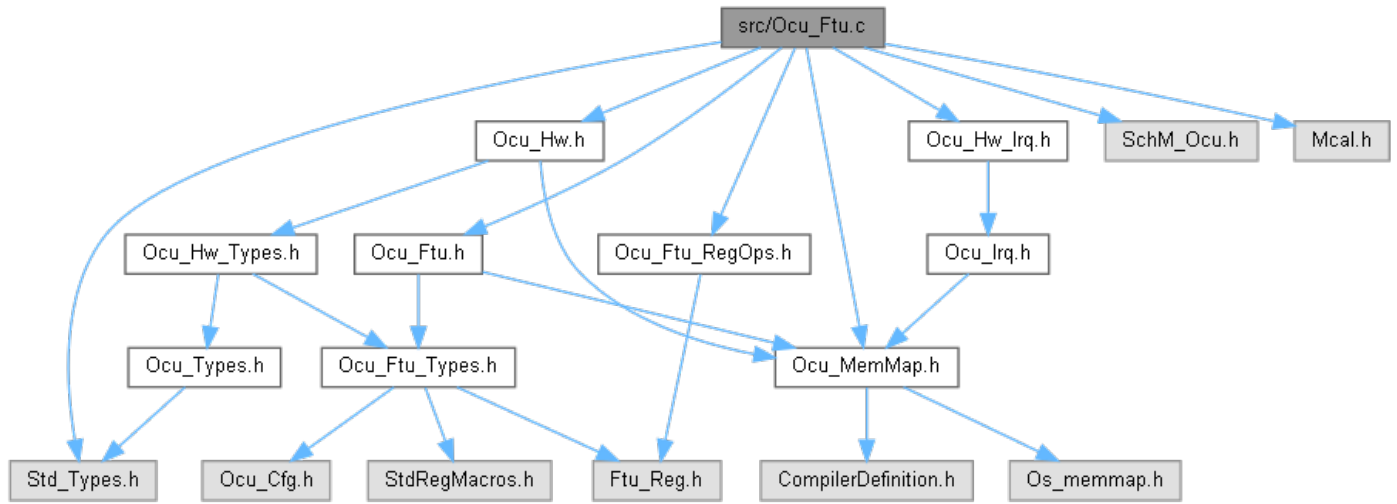
Rejected Requirement 6 **SWS_Ocu_00156**

Description	This parameter is used to allow the OCU channel to trigger a DMA channel upon compare match, if this is supported by hardware. The value of the parameter represents the DMA physical channel to trigger.
Rejection Reason	Hardware trigger for DMA has not implemented in OCU driver yet.

Rejected Requirement 6 SWS_Ocu_00168	
Description	Maps the OCU kernel to zero or one ECUC partitions to assign the driver kernel to a certain core. The ECUC partition referenced is a subset of the ECUC partitions where the OCU driver is mapped to.
Rejection Reason	Type IV Autosar multicore not implemented for current module, therefore OcuKernelEcucPartitionRef is not supported.

2.2 File Structure





2.3 Macros

2.3.1 Macros in Ocu.h

- **#define OCU_INDEX** ((uint8)0x0U)
Instance ID of this OCU driver
- **#define OCU_E_UNINIT** ((uint8)0x02U)
API service used without module initialization.
- **#define OCU_E_PARAM_INVALID_CHANNEL** ((uint8)0x03U)
API service used with an invalid channel Identifier.
- **#define OCU_E_PARAM_INVALID_STATE** ((uint8)0x04U)
AP IOcu_SetPinState service called with an invalid pin state or when the channel is in the RUNNING state.
- **#define OCU_E_PARAM_INVALID_ACTION** ((uint8)0x05U)
API Ocu_SetPinAction service called with an invalid pin action.
- **#define OCU_E_NO_VALID_NOTIF** ((uint8)0x06U)
API for notification enabling/disabling services called with a NULL function pointer.
- **#define OCU_E_ALREADY_INITIALIZED** ((uint8)0x07U)
API Ocu_Init service called while the OCU driver has already been initialised.
- **#define OCU_E_PARAM_POINTER** ((uint8)0x08U)
Generated when a NULL pointer is passed to Ocu_GetVersionInfo function.
- **#define OCU_E_BUSY** ((uint8)0x09U)
API Ocu_StartChannel service called on a channel that is in state RUNNING.

- **#define OCU_E_PARAM_NO_PIN** ((uint8)0x0AU)
API Ocu_SetPinState or Ocu_SetPinAction services called for a channel that does not have an associated output pin.
- **#define OCU_E_INIT_FAILED** ((uint8)0x0BU)
API Ocu_Init service called with wrong parameter.
- **#define OCU_E_PARAM_INVALID_VALUE** ((uint8)0x1BU)
Function called for with a compare match parameter greater than maximum supported counter value for a given channel.
- **#define OCU_INIT_ID** ((uint8)0x00U)
API service ID of Ocu_Init function.
- **#define OCU_DEINIT_ID** ((uint8)0x01U)
API service ID of Ocu_DeInit function.
- **#define OCU_STARTCHANNEL_ID** ((uint8)0x02U)
API service ID of Ocu_StartChannel function.
- **#define OCU_STOPCHANNEL_ID** ((uint8)0x03U)
API service ID of Ocu_StopChannel function.
- **#define OCU_SETPINSTATE_ID** ((uint8)0x04U)
API service ID of Ocu_SetPinState function.
- **#define OCU_SETPINACTION_ID** ((uint8)0x05U)
API service ID of Ocu_SetPinAction function.
- **#define OCU_GETCOUNTER_ID** ((uint8)0x06U)
API service ID of Ocu_GetCounter function.
- **#define OCU_SETABSOLUTETHRESHOLD_ID** ((uint8)0x07U)
API service ID of Ocu_SetAbsoluteThreshold function.
- **#define OCU_SETRELATIVETHRESHOLD_ID** ((uint8)0x08U)
API service ID of Ocu_SetRelativeThreshold function.
- **#define OCU_GETVERSIONINFO_ID** ((uint8)0x09U)
API service ID of Ocu_GetVersionInfo function.
- **#define OCU_DISABLENOTIFICATION_ID** ((uint8)0x0AU)
API service ID of Ocu_DisableNotification function.

- #define **OCU_ENABLENOTIFICATION_ID** ((uint8)0x0BU)
API service ID of Ocu_EnableNotification function.

2.3.2 Macros in Ocu_Ftu_Types.h

- #define **OCU_FTU_SWOCTRL_FORCED_VALUE_OFFSET** ((uint8)8U)
- #define **OCU_FTU_0** ((uint8)0x00)
- #define **OCU_FTU_1** ((uint8)0x01)
- #define **OCU_FTU_2** ((uint8)0x02)
- #define **OCU_FTU_3** ((uint8)0x03)
- #define **OCU_FTU_4** ((uint8)0x04)
- #define **OCU_FTU_5** ((uint8)0x05)
- #define **OCU_FTU_6** ((uint8)0x06)
- #define **OCU_FTU_7** ((uint8)0x07)
- #define **OCU_FTU_8** ((uint8)0x08)
- #define **OCU_FTU_9** ((uint8)0x09)
- #define **OCU_FTU_10** ((uint8)0x0A)
- #define **OCU_FTU_11** ((uint8)0x0B)
- #define **OCU_FTU_CH_0** ((uint8)0x00)
- #define **OCU_FTU_CH_1** ((uint8)0x01)

- `#define OCU_FTU_CH_2 ((uint8)0x02)`
- `#define OCU_FTU_CH_3 ((uint8)0x03)`
- `#define OCU_FTU_CH_4 ((uint8)0x04)`
- `#define OCU_FTU_CH_5 ((uint8)0x05)`
- `#define OCU_FTU_CH_6 ((uint8)0x06)`
- `#define OCU_FTU_CH_7 ((uint8)0x07)`

2.3.3 Macros in Ocu_Version.h

- `#define OCU_AR_RELEASE_MAJOR_VERSION 4`
- `#define OCU_AR_RELEASE_MINOR_VERSION 6`
- `#define OCU_AR_RELEASE_REVISION_VERSION 0`
- `#define OCU_MODULE_ID 125`
- `#define OCU_SW_MAJOR_VERSION 0`
- `#define OCU_SW_MINOR_VERSION 6`
- `#define OCU_SW_PATCH_VERSION 0`
- `#define OCU_VENDOR_ID 174`

2.3.4 Macros in Ocu_Cfg.h

- `#define OCU_CFG_AR_RELEASE_MAJOR_VERSION 4`
- `#define OCU_CFG_AR_RELEASE_MINOR_VERSION 6`

- **#define OCU_CFG_AR_RELEASE_REVISION_VERSION 0**
- **#define OCU_CFG_SW_MAJOR_VERSION 0**
- **#define OCU_CFG_SW_MINOR_VERSION 6**
- **#define OCU_CFG_SW_PATCH_VERSION 0**
- **#define OCU_CFG_VENDOR_ID 174**
- **#define OCU_CFG_MODULE_ID 125**
- **#define OCU_MUTILCORE_SUPPORT (STD_OFF)**
Switch for enabling the multicore support.
- **#define OCU_MAX_PARTITIONS (0U)**
Maximum number of configured partitions.
- **#define OCU_DEV_ERROR_DETECT (STD_ON)**
Switch for enabling the development error detection.
- **#define OCU_DEINIT_API (STD_ON)**
Switch to indicate that Ocu_DeInit API is supported.
- **#define OCU_SET_PIN_STATE_API (STD_ON)**
Switch to indicate that Ocu_SetPinState API is supported.
- **#define OCU_SET_PIN_ACTION_API (STD_ON)**
Switch to indicate that Ocu_SetPinAction API is supported.
- **#define OCU_GET_COUNTER_API (STD_ON)**
Switch to indicate that Ocu_GetCounter API is supported.
- **#define OCU_VERSION_INFO_API (STD_ON)**
Switch to indicate that Ocu_GetVersionInfo API is supported.
- **#define OCU_SET_ABSOLUTE_THRESHOLD_API (STD_OFF)**
Switch to indicate that Ocu_SetAbsoluteThreshold API is supported.

- **#define OCU_SET_RELATIVE_THRESHOLD_API** (STD_ON)
Switch to indicate that Ocu_SetRelativeThreshold API is supported.
- **#define OCU_NOTIFICATION_SUPPORTED** (STD_ON)
Switch to indicate that the notifications are supported.
- **#define OCU_DOWNCOUNTING_SUPPORTED** (STD_OFF)
Switch to indicate that the down counting is supported.
- **#define OCU_DISABLE_DET_REPORT_RUNTIME_ERROR_STATUS** (STD_OFF)
Switch for enabling the runtime error detection.
- **#define OCU_PRECOMPILE_SUPPORT** (STD_OFF)
Switch to indicate that the precompile is supported.
- **#define OCU_MAX_CHANNELS** (3U)
Maximum configured channels on this platform.
- **#define OCU_FTU_HW_CHANNEL_MAX** (3U)
Maximum of FTU configured channels on this platform.
- **#define OCU_FTU_HW_MODULES** ((uint8)12U)
Maximum of FTU modules on this platform.
- **#define OCU_FTU_CH_PER_MODULES** ((uint8)8U)
Maximum of FTU channels per module on this platform.
- **#define OCU_FTU_HW_CHANNELS** (uint8)(OCU_FTU_HW_MODULES * OCU_FTU_CH_PER_MODULES)
Maximum of FTU channels that can be configured as OCU on this platform.
- **#define OCU_HW_CHANNELS_NO** (OCU_FTU_HW_CHANNELS)
- **#define OCU_FTU_MODULE_CHANNELS_NO** (8U)

2.4 Enums

2.4.1 Enumerations in Ocu.h

2.4.1.1 Ocu_ChannelStatusType

Enumeration	Ocu_ChannelStatusType	
Description	This enumerated type allows the selection of channel status type.	
Values	Value	Description
	OCU_STATUS_UNINITIALIZED = 0	OCU channel status - uninitialized
	OCU_STATUS_INITIALIZED = 1	OCU channel status - initialized
	OCU_STATUS_STOPPED = 2	OCU channel status - stopped
	OCU_STATUS_RUNNING = 3	OCU channel status - running

2.4.2 Enumerations in Ocu_Types.h

2.4.2.1 Ocu_CountDirectionType

Enumeration	Ocu_CountDirectionType	
Description	This enum specifies the count direction for the whole OCU driver.	
Values	Value	Description
	OCU_UPCOUNTING = 0U	Counter in UP Counting.
	OCU_DOWNCOUNTING = 1U	Counter in DOWN Counting.

2.4.2.2 Ocu_PinStateType

Enumeration	Ocu_PinStateType	
Description	Output state of the pin linked to an OCU channel.	
Values	Value	Description
	OCU_HIGH = 0U	The pin associated to an OCU channel is in high state.
	OCU_LOW = 1U	The pin associated to an OCU channel is in low state.

2.4.2.3 Ocu_PinActionType

Enumeration	Ocu_PinActionType	
Description	Automatic action (by hardware) to be performed on a pin attached to an OCU channel.	
Values	Value	Description
	OCU_SET_HIGH = 0	The channel pin will be set HIGH upon next compare match.
	OCU_SET_LOW = 1	The channel pin will be set LOW upon next compare match.
	OCU_TOGGLE = 2	The channel pin will be set to the opposite of its current HIGH level upon next compare match.
	OCU_DISABLE = 3	The channel pin will remain at its current level upon next compare match.

2.4.2.4 Ocu_ReturnType

Enumeration	Ocu_ReturnType
Description	Return information after setting a new threshold value.

Values	Value	Description
	OCU_CM_IN_REF_INTERVAL = 0	The compare match will occur inside the current Reference Interval.
	OCU_CM_OUT_REF_INTERVAL = 1	The compare match will not occur inside the current Reference Interval.

2.4.3 Enumerations in Ocu_Ftu_Types.h

2.4.3.1 OCU_FTU_DebugModeType

Enumeration	OCU_FTU_DebugModeType	
Description	Ftu debug mode	
Values	Value	Description
	OCU_FTU_DBG_COUNTER_STOP_CH N_WORKS = 0	FTU counter stopped, channel works as function mode.
	OCU_FTU_DBG_COUNTER_STOP_CH N_FORCE_SAFE = 1	FTU counter stopped, channel output force to safe state.
	OCU_FTU_DBG_COUNTER_STOP_CH N_FROZEN = 2	FTU counter stopped, channel output is frozen.
	OCU_FTU_DBG_COUNTER_WORKS_C HN_WORKS = 3	FTU counter works as function mode, channel works as function mode.

2.4.3.2 Ocu_FtuPrescalerType

Enumeration	Ocu_FtuPrescalerType	
Description	Ftu counter prescaler.	
Values	Value	Description
	OCU_FTU_PRESCALER_DIV1 = 0	The FTU counter clock frequency is divided by 1
	OCU_FTU_PRESCALER_DIV2 = 1	The FTU counter clock frequency is divided by 2
	OCU_FTU_PRESCALER_DIV4 = 2	The FTU counter clock frequency is divided by 4
	OCU_FTU_PRESCALER_DIV8 = 3	The FTU counter clock frequency is divided by 8
	OCU_FTU_PRESCALER_DIV16 = 4	The FTU counter clock frequency is divided by 16
	OCU_FTU_PRESCALER_DIV32 = 5	The FTU counter clock frequency is divided by 32
	OCU_FTU_PRESCALER_DIV64 = 6	The FTU counter clock frequency is divided by 64
	OCU_FTU_PRESCALER_DIV128 = 7	The FTU counter clock frequency is divided by 128

2.4.3.3 Ocu_FtuModuleClkSrcType

Enumeration	Ocu_FtuModuleClkSrcType	
Description	Ftu module clock source.	
Values	Value	Description
	OCU_FTU_NO_CLK = 0	No clock selected.
	OCU_FTU_BUS_CLK = 1	FTU input clock.
	OCU_FTU_PCC_CLK = 3	External pin input clock.
	OCU_FTU_EXTERNAL_TCK0 = 4	FTU_TCLK0 pin as FTU external clock.
	OCU_FTU_EXTERNAL_TCK1 = 5	FTU_TCLK1 pin as FTU external clock.

Data Fields	- const Ocu_ChannelType nNumChannels Number of channels for a given configuration. - const Ocu_ConfigChannelType *pOcuChannelConfig Pointer to the OCU channel configuration. - const Ocu_HwConfigType HwConfig Structure of the IP Configuration data. - const Ocu_ChannelType (*HwToLogicChannelMap)[] Hardware to logic channel map table. - const uint8 * OcuChannelPartitionMappingPtr Mutilcore channel partion map table.
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2.6.2 Ocu_ChannelConfigType

Structure	Ocu_ChannelConfigType
Description	Structure that contains OCU channel notification configuration.
Diagram	N/A
Data Fields	- boolean bOcuOutputPinEnable Output pin enabled/disable. - const Ocu_NotifyType pfOcuChannelNotification The configured notification for OcuChannel.

2.6.3 Ocu_HwConfigType

Structure	Ocu_HwConfigType
Description	Structure that contains Ocu parameters. It represents a link from the logical configuration to the hardware config.
Diagram	<pre> graph BT Ocu_HwConfigType -- pFtuConfig --> Ocu_FtuConfigType Ocu_HwConfigType -- pHwChannelsConfig --> Ocu_HwChannelConfigType Ocu_FtuConfigType -- pChannelsConfig --> Ocu_Ftu_ChannelConfigType Ocu_FtuConfigType -- pModulesConfig --> Ocu_Ftu_ModuleConfigType Ocu_HwChannelConfigType --> Ocu_Ftu_ModuleConfigType </pre>
Data Fields	- const Ocu_FtuConfigType * const pFtuConfig Pointer to the FTU configuration. - const Ocu_HwChannelConfigType (*pHwChannelsConfig)[] Array of the configured hardware channel.

2.6.4 Ocu_HwChannelConfigType

Structure	Ocu_HwChannelConfigType
Description	Index in the Hw specific configuration table.

Diagram	N/A
Data Fields	const uint8 u8IdxChannelConfig Index in the Hw specific configuration table

2.6.5 Ocu_FtuConfigType

Structure	Ocu_FtuConfigType
Description	Structure that contains the Ocu general IP configuration.
Diagram	<pre> graph TD Ocu_FtuConfigType -- pChannelsConfig --> Ocu_Ftu_ChannelConfigType Ocu_FtuConfigType -- pModulesConfig --> Ocu_Ftu_ModuleConfigType </pre>
Data Fields	- const Ocu_Ftu_ChannelType u8NumChannels; Number of Ftu channels in the Ocu configuration. - const Ocu_Ftu_ModuleType u8NumModules; Number of Ftu channels in the Ocu configuration. - const Ocu_Ftu_ChannelConfigType (*pChannelsConfig)[] List of the hardware channel configurations. - const Ocu_Ftu_ModuleConfigType (*pModulesConfig)[] List of the hardware module configurations.

2.6.6 Ocu_Ftu_ChannelConfigType

Structure	Ocu_Ftu_ChannelConfigType
Description	Structure that contains the Ocu general IP configuration.
Diagram	N/A
Data Fields	- boolean bOcuOutputPinUsed Output Pin is used. - const Ocu_PinStateType eDefaultPinState; Default Pin state of channel. - const Ocu_PinActionType eOutputPinAction OCU Pin action. - const Ocu_Ftu_ChannelType u8HwChannel Assigned Ftu channel id. - const Ocu_Ftu_ModuleType u8HwModule Assigned Ftu channel id. - const Ocu_ValueType u32DefaultThreshold Compare match threshold for the current channel.

2.6.7 Ocu_Ftu_ModuleConfigType

Structure	Ocu_Ftu_ModuleConfigType
Description	Ftu IP module specific configuration structure for the Ocu functionality.

Diagram	N/A
Data Fields	• <code>const Ocu_FtuModuleClkSrcType eClkSrc</code> FTU clock source. • <code>const Ocu_FtuPrescalerType ePrescaler</code> Clock prescale. • <code>const OCU_FTU_DebugModeType eDbgMode</code> Debug mode config. • <code>const uint8 u8ModuleId</code> Assigned Ftu module id. • <code>const uint32 u32MaxCounterValue</code> Maximum counter value.

2.6.8 Ocu_SetThresholdType

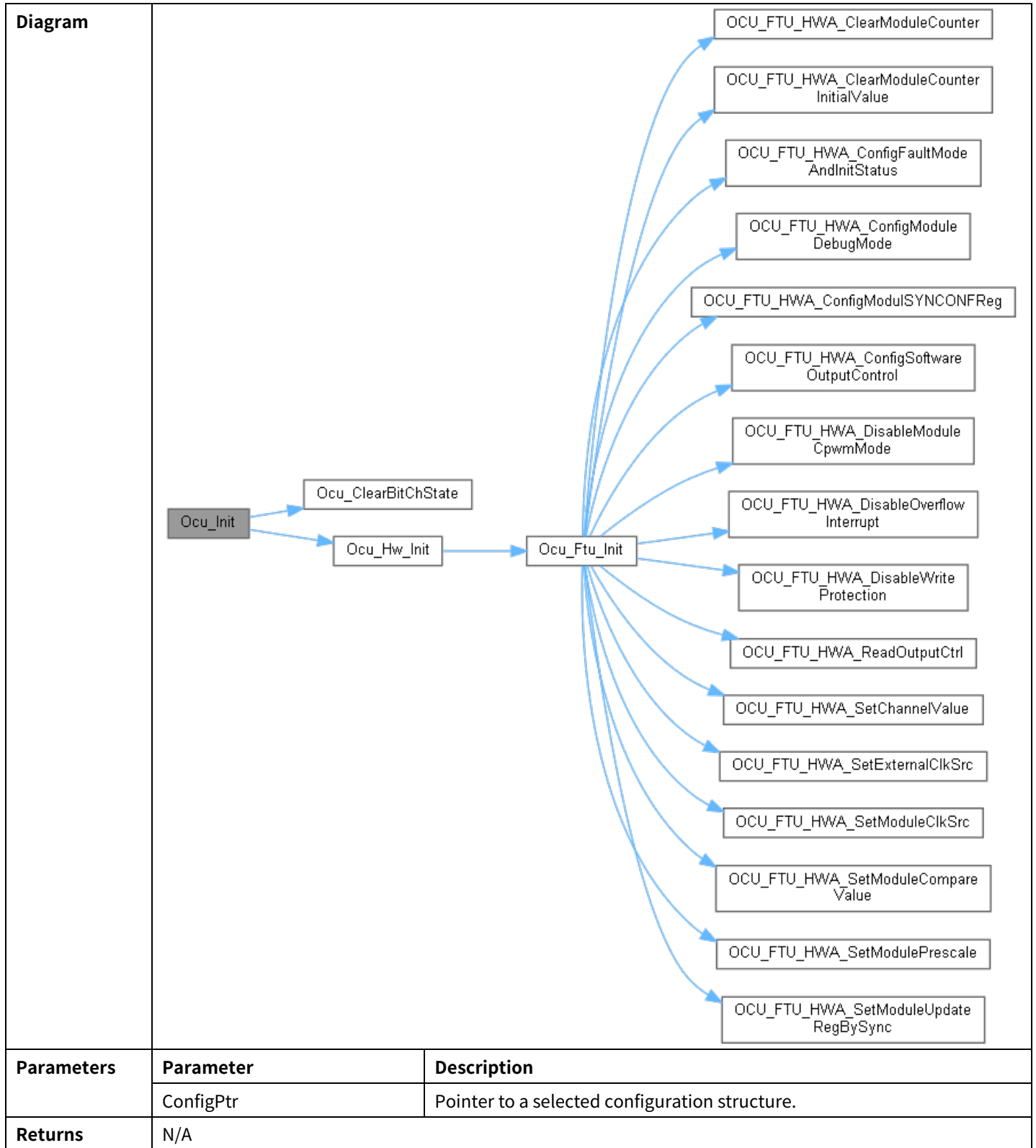
Structure	<code>Ocu_SetThresholdType</code>
Description	IP specific structure for OCU SetRelativeThreshold and SetAbsoluteThreshold functionalities.
Diagram	N/A
Data Fields	• <code>Ocu_ChannelType u8ChannelNumber</code> Channel id. • <code>Ocu_ValueType u32ReferenceValue</code> Reference interval. • <code>Ocu_ValueType u32AbsoluteValue</code> Absolute threshold value • <code>Ocu_ValueType u32RelativeValue</code> Relative threshold value • <code>Ocu_ReturnType eInterval</code> Returned status

2.7 API Functions

2.7.1 Functions in Ocu.h

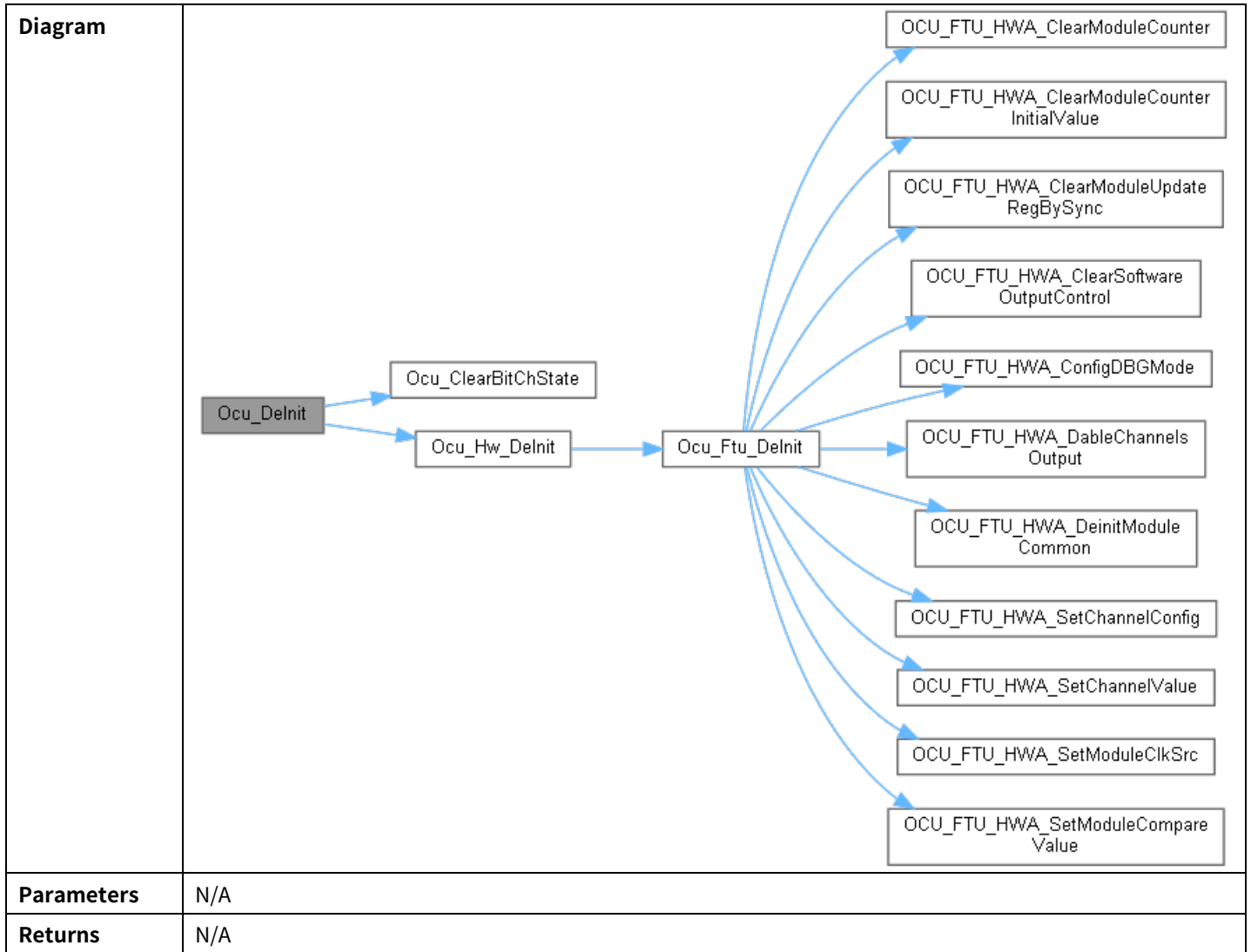
2.7.1.1 `void Ocu_Init (const Ocu_ConfigType* ConfigPtr)`

Function	<code>void Ocu_Init (const Ocu_ConfigType * ConfigPtr)</code>
Description	Initialize the OCU driver.

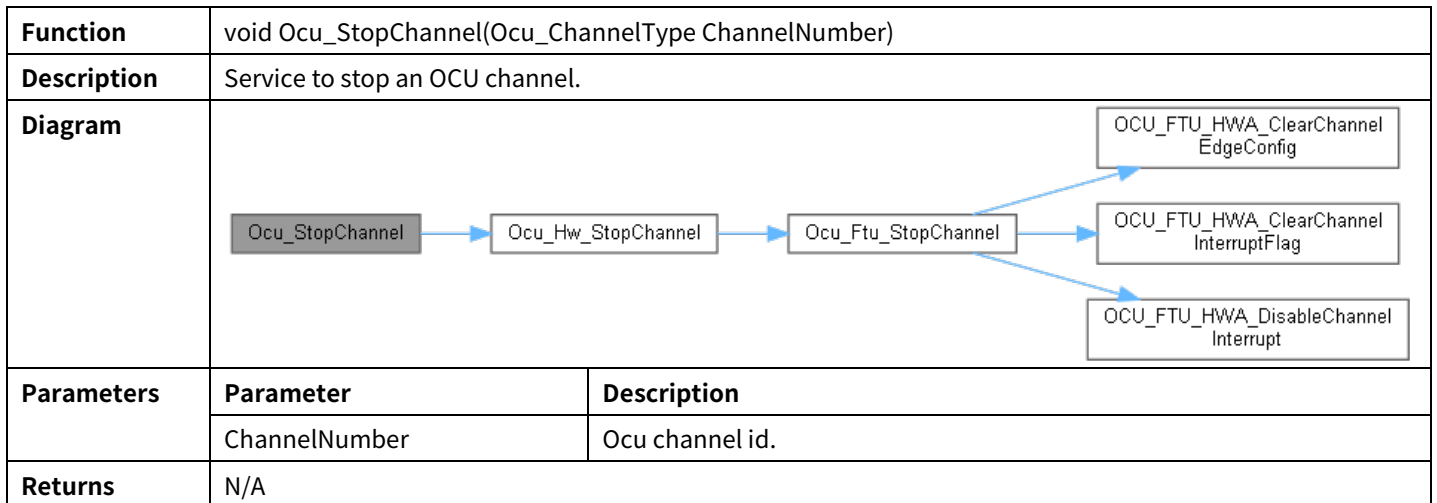


2.7.1.2 void Ocu_Delnit (void)

Function	void Ocu_Delnit (void)
Description	De-initialization.



2.7.1.3 void Ocu_StopChannel(Ocu_ChannelType ChannelNumber)



2.7.1.4 Std_ReturnType Ocu_StartChannel(Ocu_ChannelType ChannelNumber)

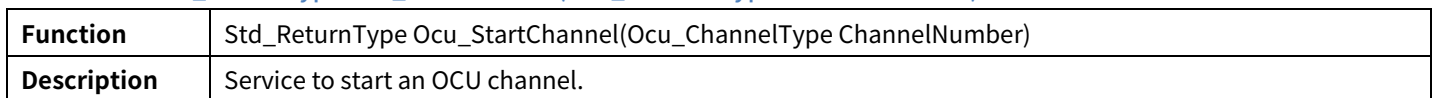


Diagram	<pre>graph LR; Ocu_StartChannel[Ocu_StartChannel] --> Ocu_Hw_StartChannel[Ocu_Hw_StartChannel]; Ocu_Hw_StartChannel --> Ocu_Ftu_StartChannel[Ocu_Ftu_StartChannel]; Ocu_Ftu_StartChannel --> OCU_FTU_HWA_ClearChannelInterruptFlag[OCU_FTU_HWA_ClearChannelInterruptFlag]; Ocu_Ftu_StartChannel --> OCU_FTU_HWA_ClearSoftwareOutputControl[OCU_FTU_HWA_ClearSoftwareOutputControl]; Ocu_Ftu_StartChannel --> OCU_FTU_HWA_EnableChannelInterrupt[OCU_FTU_HWA_EnableChannelInterrupt]; Ocu_Ftu_StartChannel --> OCU_FTU_HWA_EnableChannelsOutput[OCU_FTU_HWA_EnableChannelsOutput]; Ocu_Ftu_StartChannel --> OCU_FTU_HWA_ReadChannelInterruptFlag[OCU_FTU_HWA_ReadChannelInterruptFlag];</pre>					
Parameters	<table><tr><th>Parameter</th><th>Description</th></tr><tr><td>ChannelNumber</td><td>Ocu channel id.</td></tr></table>	Parameter	Description	ChannelNumber	Ocu channel id.	
Parameter	Description					
ChannelNumber	Ocu channel id.					
Returns	Std_ReturnType -E_OK Channel was started, E_NOT_OK Channel was not started					

2.7.1.5 void Ocu_SetPinState(Ocu_ChannelType ChannelNumber, Ocu_PinStateType PinState)

Function	void Ocu_SetPinState(Ocu_ChannelType ChannelNumber, Ocu_PinStateType PinState)	
Description	Enable channel notification.	
Diagram	<pre>graph LR; A[Ocu_SetPinState] --> B[Ocu_Hw_SetPinState]; B --> C[Ocu_Ftu_SetPinState]; C --> D[OCU_FTU_HWA_ConfigSoftwareOutputControl]; C --> E[OCU_FTU_HWA_ReadOutputCtrl];</pre>	
Parameters	Parameter	Description
	ChannelNumber	Ocu channel id.
	PinState	Output Pin State.
Returns	N/A	

2.7.1.6 void Ocu_SetPinAction(Ocu_ChannelType ChannelNumber, Ocu_PinActionType PinAction)

Function	void Ocu_SetPinAction(Ocu_ChannelType ChannelNumber, Ocu_PinActionType PinAction)	
Description	Enable the channel wakeup.	
Diagram	<pre>graph LR; A[Ocu_SetPinAction] --> B[Ocu_Hw_SetPinAction]; B --> C[Ocu_Ftu_SetPinAction];</pre>	
Parameters	Parameter	Description
	ChannelNumber	Ocu channel id.
	PinAction	OCU_SET_LOW, OCU_SET_HIGH, OCU_TOGGLE, OCU_DISABLE.
Returns	N/A	

2.7.1.7 Ocu_ValueType Ocu_GetCounter(Ocu_ChannelType ChannelNumber)

Function	Ocu_ValueType Ocu_GetCounter(Ocu_ChannelType ChannelNumber)		
Description	Service to read the current value of the counter.		
Diagram			
Parameters	Parameter	Description	
	ChannelNumber	Ocu channel id.	
Returns	Ocu_ValueType -Content of the counter in ticks.		

2.7.1.8 Ocu_ReturnType Ocu_SetAbsoluteThreshold(Ocu_ChannelType ChannelNumber, Ocu_ValueType ReferenceValue, Ocu_ValueType AbsoluteValue)

Function	Ocu_ReturnType Ocu_SetAbsoluteThreshold(Ocu_ChannelType ChannelNumber, Ocu_ValueType ReferenceValue, Ocu_ValueType AbsoluteValue)	
Description	Service to set the value of the channel threshold using an absolute input data.	
Diagram	<pre> graph LR A[Ocu_SetAbsoluteThreshold] --> B[Ocu_Hw_GetMaxCountValue] A --> C[Ocu_Hw_SetAbsoluteThreshold] B --> D[Ocu_Ftu_GetMaxCountValue] C --> E[Ocu_Ftu_SetAbsoluteThreshold] D --> F[OCU_FTU_HWA_ReadModuleCounter] E --> G[OCU_FTU_HWA_SetChannelValue] </pre>	
Parameters	Parameter	Description
	ChannelNumber	Ocu channel id.
	ReferenceValue	Value given by the upper layer and used as a base to determine whether to call the notification before the function exits or not.
	AbsoluteValue	Value to compare with the content of the counter. This value is in ticks
Returns	Ocu_ReturnType -Tells the caller whether the compare match will occur (or has already occurred) during the current Reference Interval, as a result of setting the new threshold value.	

2.7.1.9 Ocu_ReturnType Ocu_SetRelativeThreshold(Ocu_ChannelType ChannelNumber, Ocu_ValueType RelativeValue)

Function	Ocu_ReturnType Ocu_SetRelativeThreshold(Ocu_ChannelType ChannelNumber, Ocu_ValueType RelativeValue)	
Description	Service to set the value of the channel threshold relative to the current value of the counter.	
Diagram	<pre> graph LR A[Ocu_SetRelativeThreshold] --> B[Ocu_Hw_SetRelativeThreshold] B --> C[Ocu_Ftu_SetRelativeThreshold] C --> D[OCU_FTU_HWA_ReadModuleCounter] C --> E[OCU_FTU_HWA_SetChannelValue] </pre>	
Parameters	Parameter	Description
	ChannelNumber	Ocu channel id.
	RelativeValue	Value to use for computing the new threshold.
Returns	Ocu_ReturnType -Tells the caller whether the compare match will occur (or has already occurred) during the current Reference Interval, as a result of setting the new threshold value.	

2.7.1.10 void Ocu_DisableNotification(Ocu_ChannelType ChannelNumber)

Function	void Ocu_DisableNotification(Ocu_ChannelType ChannelNumber)	
Description	This service is used to disable notifications from an OCU channel.	
Diagram	<pre> graph LR A[Ocu_DisableNotification] --> B[Ocu_ClearBitChState] </pre>	
Parameters	Parameter	Description
	ChannelNumber	Ocu channel id.
Returns	N/A	

2.7.1.11 void Ocu_EnableNotification(Ocu_ChannelType ChannelNumber)

Function	void Ocu_EnableNotification(Ocu_ChannelType ChannelNumber)	
Description	This service is used to enable notifications from an OCU channel.	
Diagram	<pre> graph LR A[Ocu_EnableNotification] --> B[Ocu_SetBitChState] </pre>	

Parameters	Parameter	Description
	ChannelNumber	Ocu channel id.
Returns	N/A	

2.7.1.12 void Ocu_GetVersionInfo(Std_VersionInfoType * versioninfo)

Function	void Ocu_GetVersionInfo(Std_VersionInfoType * versioninfo)	
Description	This function returns Ocu driver version details.	
Diagram	N/A	
Parameters	Parameter	Description
	versioninfo	pointer to Std_VersionInfoType output variable.
Returns	N/A	

2.8 Hardware Functions

2.8.1 Functions in Ocu_Hw.h

2.8.1.1 void Ocu_Hw_Init(const Ocu_HwConfigType * pHwConfig, const uint8 u8Channel)

Function	void Ocu_Hw_Init(const Ocu_HwConfigType * pHwConfig, const uint8 u8Channel)	
Description	Initialize the OCU driver.	
Parameters	Parameter	Description
	pHwConfig	Pointer to module configuration structure
	u8Channel	Channel index.
Returns	N/A	
Referenced By	Ocu_Init()	

2.8.1.2 void Ocu_Hw_DeInit(const Ocu_HwConfigType * pHwConfig, const uint8 u8Channel)

Function	void Ocu_Hw_DeInit(const Ocu_HwConfigType * pHwConfig, const uint8 u8Channel)	
Description	De-initialization.	
Parameters	Parameter	Description
	pHwConfig	Pointer to module configuration structure
	u8Channel	Channel index.
Returns	N/A	
Referenced By	Ocu_DeInit()	

2.8.1.3 void Ocu_Hw_StartChannel(Ocu_ChannelType u8ChannelNumber)

Function	void Ocu_Hw_StartChannel(Ocu_ChannelType u8ChannelNumber)	
Description	Start a hardware OCU channel	
Parameters	Parameter	Description
	u8ChannelNumber	Ocu channel id in top configuration structure.
Returns	N/A	
Referenced By	Ocu_StartChannel().	

2.8.1.4 void Ocu_Hw_StopChannel(Ocu_ChannelType u8ChannelNumber)

Function	void Ocu_Hw_StopChannel(Ocu_ChannelType u8ChannelNumber)	
Description	Stop a hardware OCU channel	
Parameters	Parameter	Description
	u8ChannelNumber	Ocu channel id in top configuration structure.
Returns	N/A	
Referenced By	Ocu_StopChannel().	

2.8.1.5 void Ocu_Hw_SetPinState(Ocu_ChannelType u8ChannelNumber, Ocu_PinStateType ePinState)

Function	void Ocu_Hw_SetPinState(Ocu_ChannelType u8ChannelNumber, Ocu_PinStateType ePinState)	
Description	Set OCU hardware channel pin state.	
Parameters	Parameter	Description
	u8ChannelNumber	Ocu channel id in top configuration structure.
	ePinState	Expected Pin State for the given channel
Returns	N/A	
Referenced By	Ocu_SetPinState()	

2.8.1.6 void Ocu_Hw_SetPinAction(Ocu_ChannelType u8ChannelNumber, Ocu_PinActionType ePinAction)

Function	void Ocu_Hw_SetPinAction(Ocu_ChannelType u8ChannelNumber, Ocu_PinActionType ePinAction)	
Description	Hardware level set Pin action.	
Parameters	Parameter	Description
	u8ChannelNumber	Ocu channel id in top configuration structure.
	ePinAction	Expected Pin Action for the given channel
Returns	N/A	
Referenced By	Ocu_SetPinAction()	

2.8.1.7 Ocu_ValueType Ocu_Hw_GetCounter(Ocu_ChannelType u8ChannelNumber)

Function	Ocu_ValueType Ocu_Hw_GetCounter(Ocu_ChannelType u8ChannelNumber)	
Description	Hardware level get timer counter value.	
Parameters	Parameter	Description
	u8ChannelNumber	Ocu channel id in top configuration structure
Returns	Ocu_ValueType -counter value	
Referenced By	Ocu_GetCounter()	

2.8.1.8 Ocu_ReturnType Ocu_Hw_SetAbsoluteThreshold(Ocu_ChannelType u8ChannelNumber, Ocu_ValueType ReferenceValue, Ocu_ValueType AbsoluteValue)

Function	Ocu_ReturnType Ocu_Hw_SetAbsoluteThreshold(Ocu_ChannelType u8ChannelNumber, Ocu_ValueType ReferenceValue, Ocu_ValueType AbsoluteValue)	
Description	Hardware level set absolute threshold	
Parameters	Parameter	Description
	u8ChannelNumber	Ocu channel id in top configuration structure

	ReferenceValue	Reference value for the counting interval
	AbsoluteValue	Absolute value for the counting interval
Returns	Ocu_ReturnType	
Referenced By	Ocu_SetAbsoluteThreshold()	

2.8.1.9 Ocu_ReturnType Ocu_Hw_SetRelativeThreshold(Ocu_ChannelType u8ChannelNumber, Ocu_ValueType RelativeValue)

Function	Ocu_ReturnType Ocu_Hw_SetRelativeThreshold(Ocu_ChannelType u8ChannelNumber, Ocu_ValueType RelativeValue)	
Description	Hardware level set relative threshold.	
Parameters	Parameter	Description
	u8ChannelNumber	Ocu channel id in top configuration structure
	RelativeValue	Reference value for the counting interval
Returns	Ocu_ReturnType	
Referenced By	Ocu_SetRelativeThreshold()	

2.8.2 Functions in Ocu_Ftu.h

2.8.2.1 void Ocu_Ftu_Init(const Ocu_Ftu_ChannelConfigType * pFtuChannelConfig, const Ocu_Ftu_ModuleConfigType * pFtuModuleConfig, const uint8 u8Channel)

Function	void Ocu_Ftu_Init(const Ocu_Ftu_ChannelConfigType * pFtuChannelConfig, const Ocu_Ftu_ModuleConfigType * pFtuModuleConfig, const uint8 u8Channel)	
Description	Initialize FTU Modules and channels	
Parameters	Parameter	Description
	pFtuChannelConfig	Pointer to FTU channel configuration structure.
	pFtuModuleConfig	Pointer to FTU channel configuration structure.
	u8Channel	Channel index uses to store channel state.
Returns	N/A	
Referenced By	Ocu_Hw_Init()	

2.8.2.2 void Ocu_Ftu_Delnit(const Ocu_Ftu_ChannelConfigType * pFtuChannelConfig, const Ocu_Ftu_ModuleConfigType * pFtuModuleConfig, const uint8 u8Channel)

Function	void Ocu_Ftu_Delnit(const Ocu_Ftu_ChannelConfigType * pFtuChannelConfig, const Ocu_Ftu_ModuleConfigType * pFtuModuleConfig, const uint8 u8Channel)	
Description	De-initialization.	
Parameters	Parameter	Description
	pFtuChannelConfig	Pointer to FTU channel configuration structure.
	pFtuModuleConfig	Pointer to FTU channel configuration structure.
	u8Channel	Channel index uses to store channel state.
Returns	N/A	
Referenced By	Ocu_Hw_Delnit()	

2.8.2.3 void Ocu_Ftu_StartChannel(uint8 u8ChannelIdx, uint8 u8HwModule, uint8 u8HwChannel)

Function	void Ocu_Ftu_StartChannel(uint8 u8ChannelIdx, uint8 u8HwModule, uint8 u8HwChannel)	
Description	Get the predef timer value.	
Parameters	Parameter	Description
	u8ChannelIdx	FTU channel id.
	u8HwModule	FTU module hardware number.
	u8HwChannel	FTU channel hardware number.
Returns	N/A	
Referenced By	Ocu_Hw_StartChannel()	

2.8.2.4 void Ocu_Ftu_StopChannel(uint8 u8HwModule, uint8 u8HwChannel)

Function	void Ocu_Ftu_StopChannel(uint8 u8HwModule, uint8 u8HwChannel)	
Description	Stop FTU timer channel.	
Parameters	Parameter	Description
	u8HwModule	FTU module hardware number.
	u8HwChannel	FTU channel hardware number.
Returns	N/A	
Referenced By	Ocu_Hw_StopChannel()	

2.8.2.5 void Ocu_Ftu_SetPinState(uint8 u8HwModule, uint8 u8HwChannel, Ocu_PinStateType ePinState)

Function	void Ocu_Ftu_SetPinState(uint8 u8HwModule, uint8 u8HwChannel, Ocu_PinStateType ePinState)	
Description	Ocu driver function for setting the Pin State for the current Ftu channel.	
Parameters	Parameter	Description
	u8HwModule	FTU module hardware number.
	u8HwChannel	FTU channel hardware number.
	ePinState	State of pin.
Returns	N/A	
Referenced By	Ocu_Hw_SetPinState()	

2.8.2.6 void Ocu_Ftu_SetPinAction(const uint8 u8ChannelIndex, Ocu_PinActionType ePinAction, const Ocu_FtuConfigType * const pConfig)

Function	void Ocu_Ftu_SetPinAction(const uint8 u8ChannelIndex, Ocu_PinActionType ePinAction, const Ocu_FtuConfigType * const pConfig)	
Description	This function changes the output pin behaviour of the selected channel to the one specified for next comparison match.	
Parameters	Parameter	Description
	u8ChannelIndex	FTU channel id.
	ePinAction	OCU_SET_LOW, OCU_SET_HIGH, OCU_TOGGLE, OCU_DISABLE.
	pConfig	Pointer to Ftu hardware configuration structure.
Returns	N/A	
Referenced By	Ocu_Hw_SetPinAction()	

2.8.2.7 uint16 Ocu_Ftu_GetCounter(uint8 u8HwModule)

Function	Uint32 Ocu_Ftu_GetCounter(uint8 u8HwModule)	
Description	Ocu driver function reads and sends back the counter value of the selected channel.	
Parameters	Parameter	Description
	u8HwModule	FTU module hardware number.
Returns	The current value of CNT register.	
Referenced By	Ocu_Hw_GetCounter()	

2.8.2.8 void Ocu_Ftu_SetAbsoluteThreshold(Ocu_SetThresholdType * SetThreshold, const Ocu_FtuConfigType * const pFtuConfig)

Function	void Ocu_Ftu_SetAbsoluteThreshold(Ocu_SetThresholdType * SetThreshold, const Ocu_FtuConfigType * const pFtuConfig)	
Description	This function sets the comparison threshold of the selected channel using an absolute input data.	
Parameters	Parameter	Description
	SetThreshold	Struct contain u8ChannelNumber, ReferenceValue, AbsoluteValue and Ocu_ReturnType.
	pFtuConfig	Pointer to Ftu hardware configuration structure.
Returns	N/A	
Referenced By	Ocu_Hw_SetAbsoluteThreshold()	

2.8.2.9 void Ocu_Ftu_SetRelativeThreshold(Ocu_SetThresholdType * SetThreshold, const Ocu_FtuConfigType * const pFtuConfig)

Function	void Ocu_Ftu_SetRelativeThreshold(Ocu_SetThresholdType * SetThreshold, const Ocu_FtuConfigType * const pFtuConfig)	
Description	The function sets the value of the channel threshold relative to the current value of the counter.	
Parameters	Parameter	Description
	SetThreshold	Struct contain u8ChannelNumber, ReferenceValue, AbsoluteValue and Ocu_ReturnType.
	pFtuConfig	Pointer to Ftu hardware configuration structure.
Returns	N/A	
Referenced By	Ocu_HL_StopTimer()	

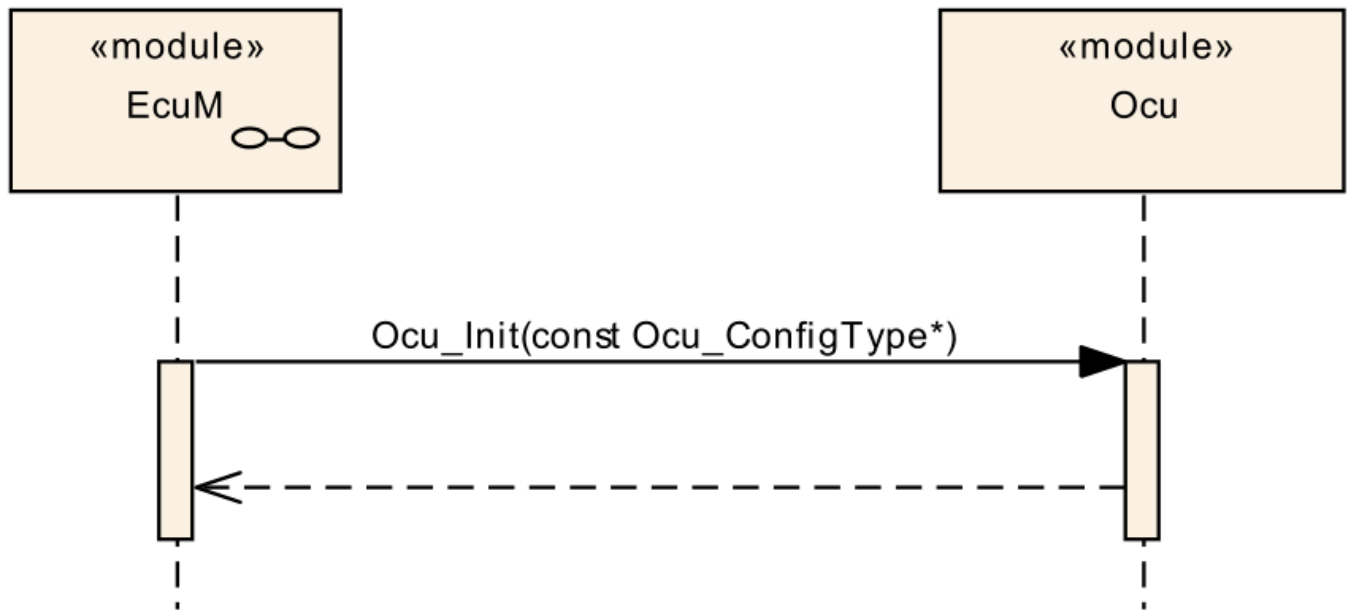
2.9 API Sequence Diagram**2.9.1 Initialization**

Ocu driver initialization.

Used HW is configured according to referenced configuration structure.

- a) All notifications are disabled.

- b) All used interrupts are disabled.

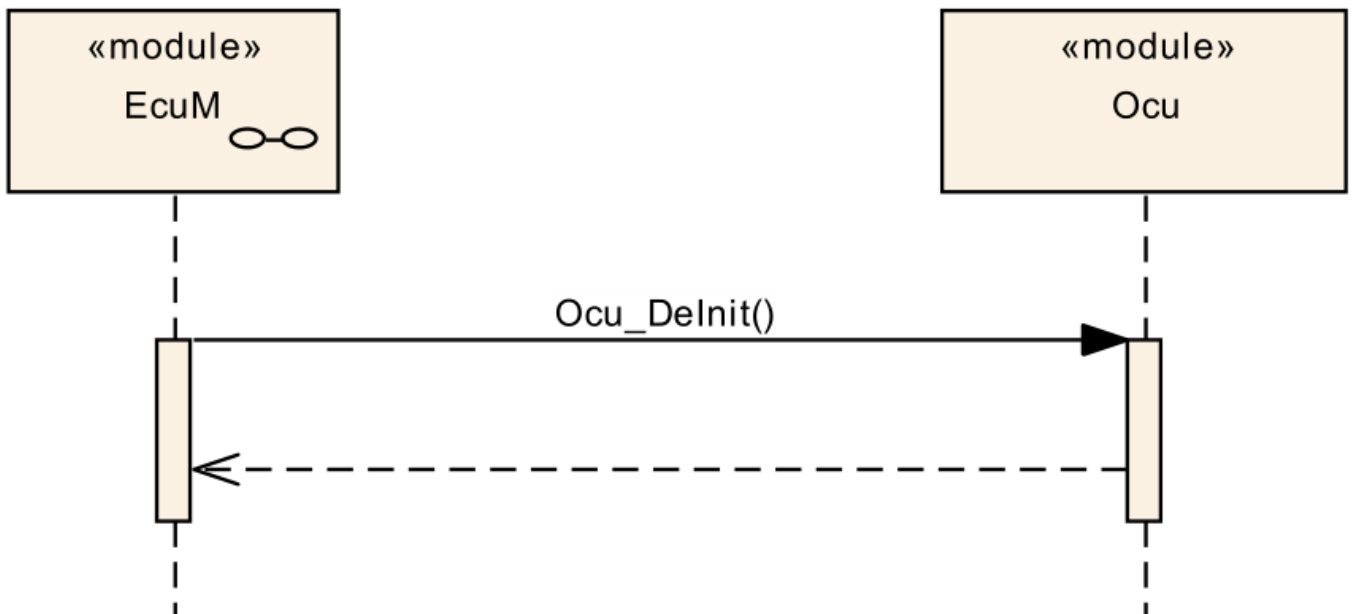


2.9.2 De-initialization

Ocu driver de-initialization.

Used HW is configured according to referenced configuration structure.

- a) All notifications are disabled.
b) All used interrupts are disabled.

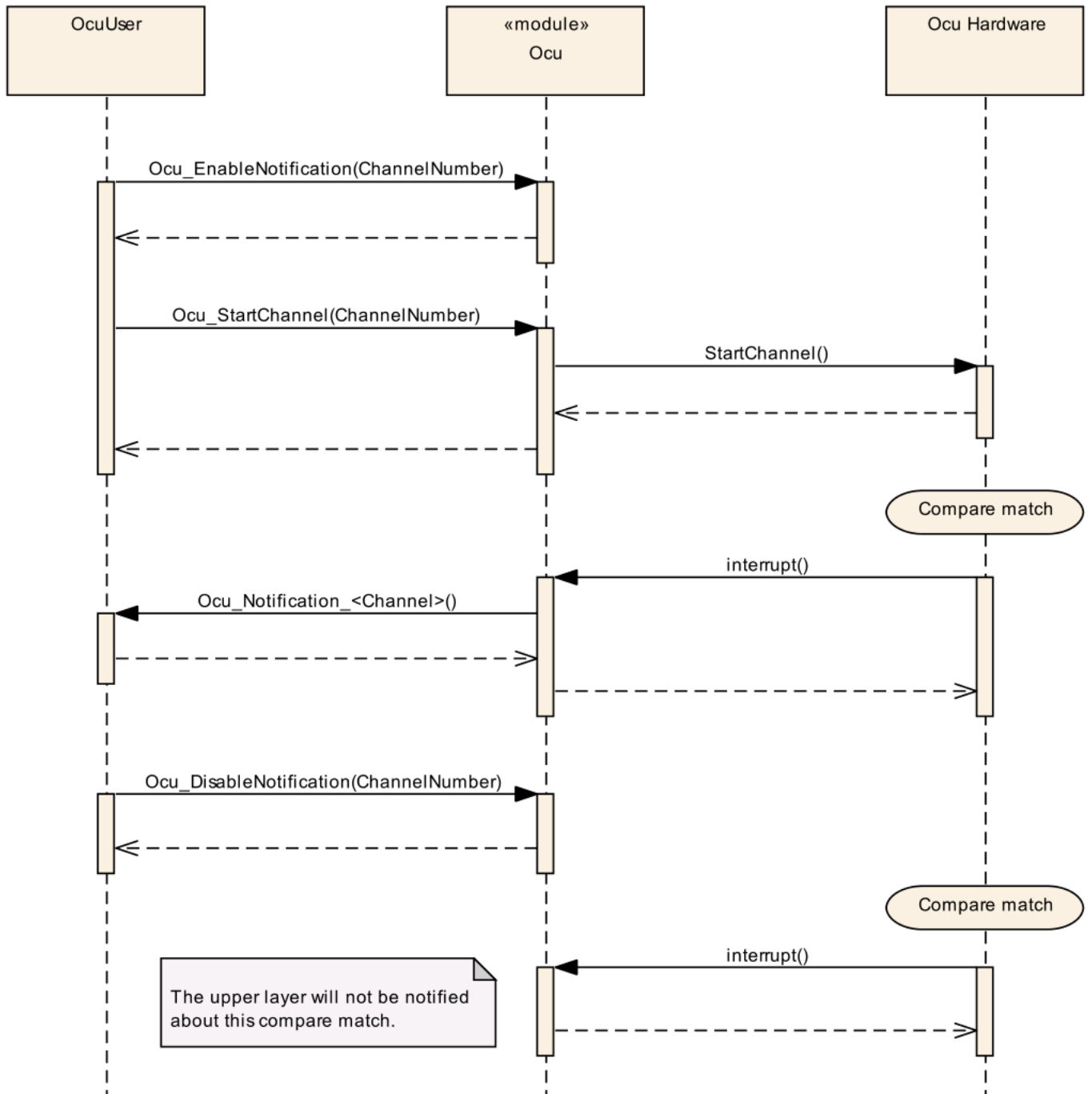


2.9.3 Using the Ocu Notifications

Using OCU Enable/Disable notification.

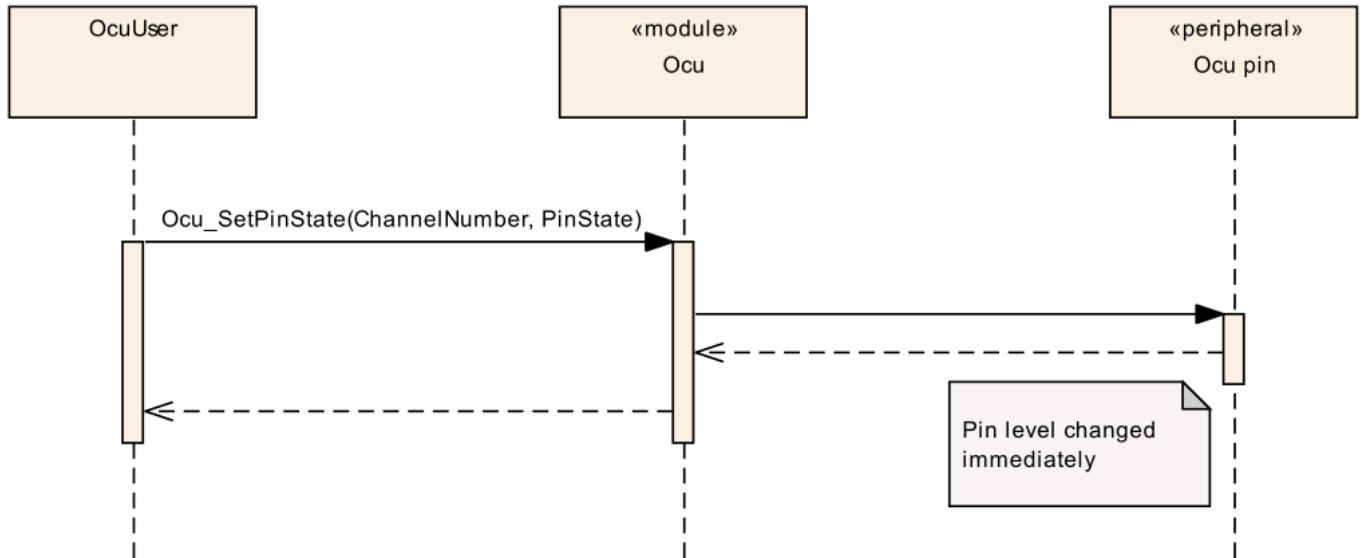
The OCU channel is started after `Ocu_StartChannel()`. No notifications occur until the first call of

`Ocu_EnableNotification(...)`. `Ocu_Notification<#Channel>` represents the configured callback function for a channel.



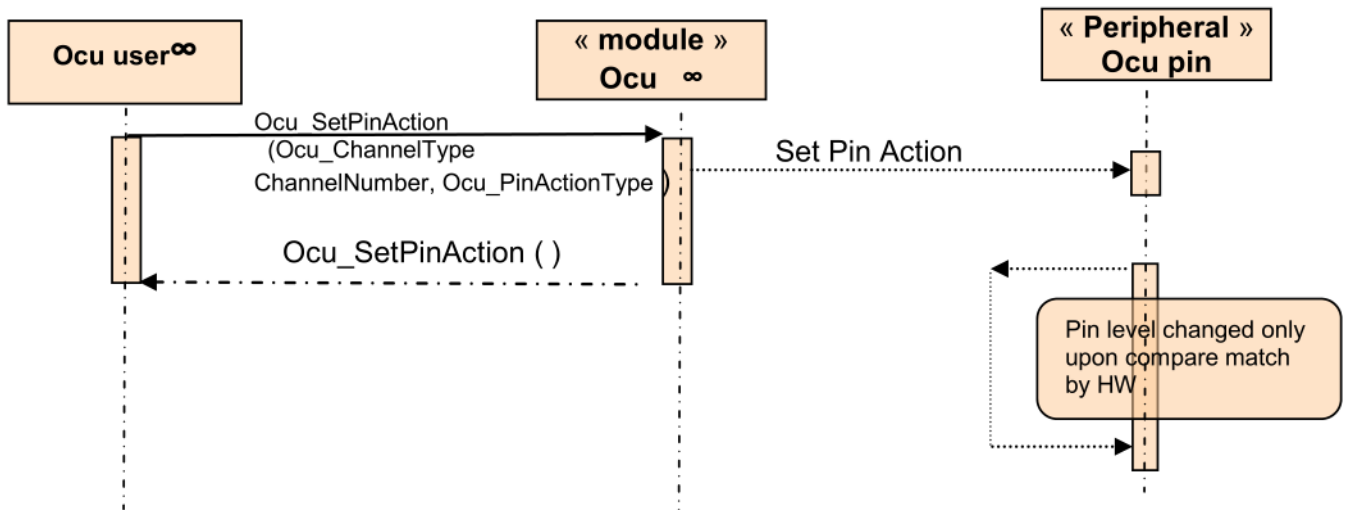
2.9.4 Ocu_SetPinState

Set Ocu output pin status. The pin level will be changed immediately after calling `Ocu_SetPinState()`.



2.9.5 Ocu_SetPinAction

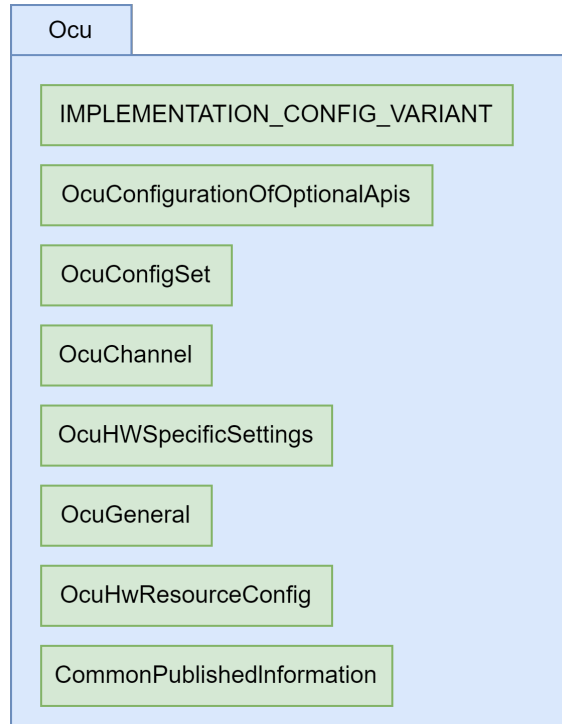
Set Ocu output pin action. After calling the Ocu_SetPinAction(), the pin level only changes on the compare match.



Chapter 3 Tresos Configuration Items

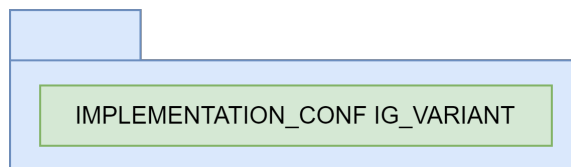
3.1 Container Inclusion Relation

The OCU module has the following 7 containers:

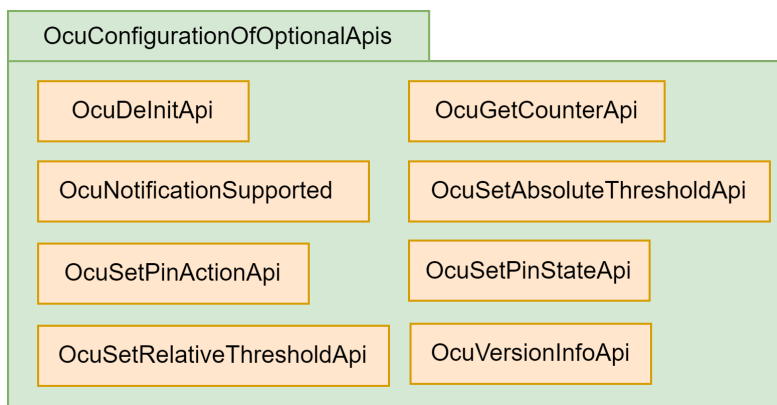


And the content of each container is as follows:

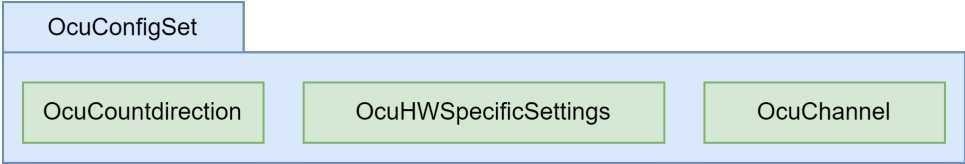
3.1.1 IMPLEMENTATION_CONFIG_VARIANT



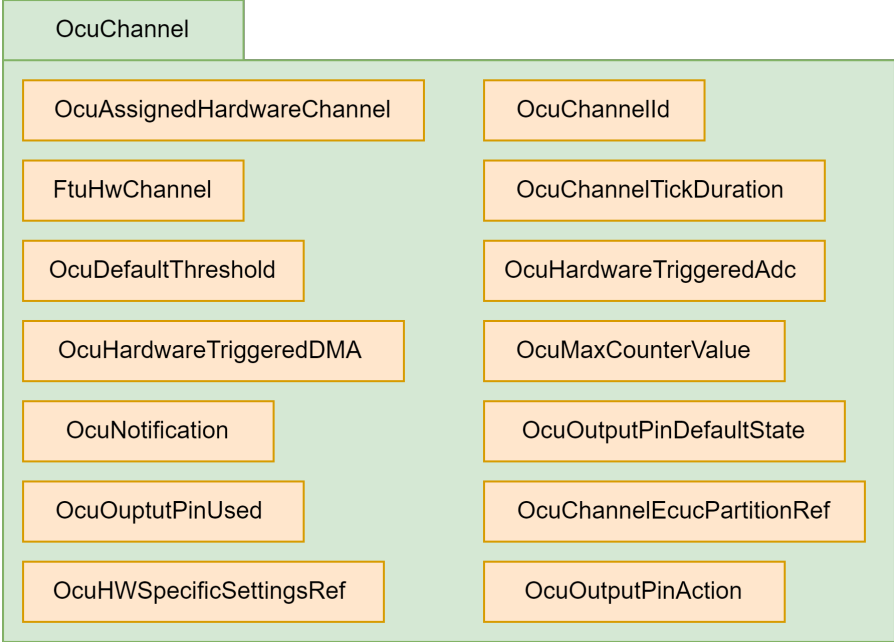
3.1.2 OcuConfigurationOfOptionalApis



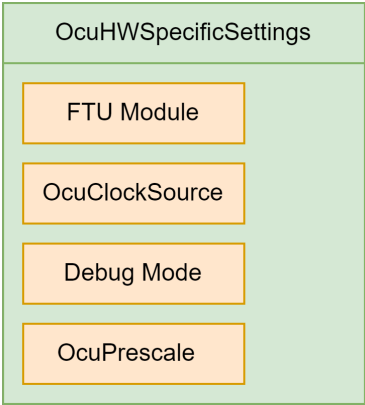
3.1.3 OcuConfigSet



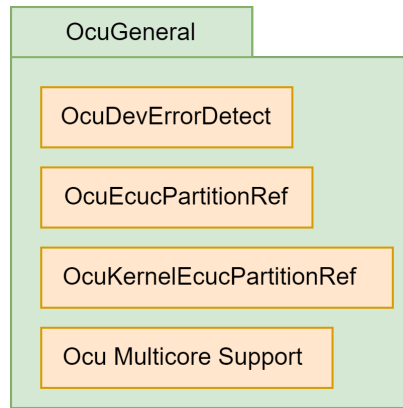
3.1.4 OcuChannel



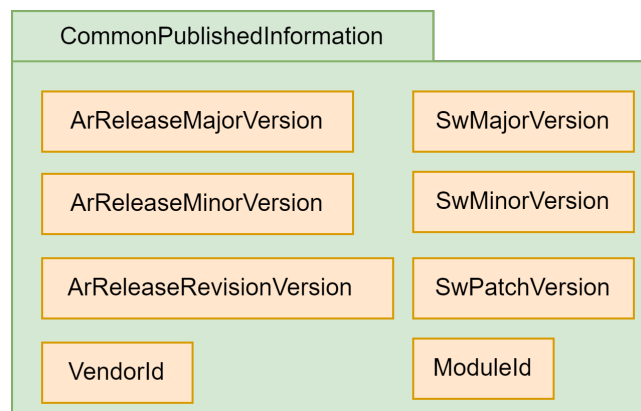
3.1.5 OcuHWSpecificSettings



3.1.6 OcuGeneral



3.1.7 CommonPublishedInformation



3.2 Containers and Variables

3.2.1 IMPLEMENTATION_CONFIG_VARIANT

Container	IMPLEMENTATION_CONFIG_VARIANT	
Description	N/A	
Screenshot		
Properties	Property	Value
	Type	Variable: ENUMERATION
	Label	Config Variant
	Default	VariantPostBuild
	Range	VariantPostBuild VariantPreCompile
Returns	N/A	

3.2.2 OcuConfigurationOfOptionalApis

Container	OcuConfigurationOfOptionalApis
Description	This container contains all configuration switches for configuring optional API services of the OCU driver.

Screenshot	▼ OcuConfigurationOfOptionalApis Name  OcuConfigurationOfOptionalApis OcuDeInitApi  OcuGetCounterApi  OcuNotificationSupported  OcuSetAbsoluteThresholdApi  OcuSetPinActionApi  OcuSetPinStateApi  OcuSetRelativeThresholdApi  OcuVersionInfoApi 				
	Properties	<table><tr><th>Property</th><th>Value</th></tr><tr><td>Type</td><td>Container</td></tr></table>	Property	Value	Type
Property	Value				
Type	Container				

3.2.2.1 OcuDeinitApi

Variable	OcuDeinitApi	
Description	Add/Remove the service Ocu_DeInit() from the code.	
Screenshot	OcuDeInitApi 📄 ☑ 🔗 ▼	
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false

3.2.2.2 OcuGetCounterApi

Variable	OcuGetCounterApi	
Description	Add/Remove the services Ocu_GetCounter() from the code.	
Screenshot	OcuGetCounterApi 📄 ☑ 🔗 ▼	
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false

3.2.2.3 OcuNotificationSupported

Variable	OcuNotificationSupported	
Description	Adds/removes the services Ocu_EnableNotification() and Ocu_DisableNotification() from the code.	
Screenshot	OcuNotificationSupported 📄 ☑ 🔗 ▼	
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false

3.2.2.4 OcuSetAbsoluteThresholdApi

Variable	OcuSetAbsoluteThresholdApi	
Description	Adds/removes the service Ocu_SetAbsoluteThreshold() from the code.	

Screenshot	OcuSetAbsoluteThresholdApi    ▼	
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False

3.2.2.5 [OcuSetPinActionApi](#)

Variable	OcuSetPinActionApi	
Description	Adds/removes the service Ocu_SetPinAction() from the code.	
Screenshot	OcuSetPinActionApi    ▼	
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false

3.2.2.6 [OcuSetPinStateApi](#)

Variable	OcuSetPinStateApi	
Description	Adds/removes the service Ocu_SetPinState() from the code.	
Screenshot	OcuSetPinStateApi    ▼	
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false

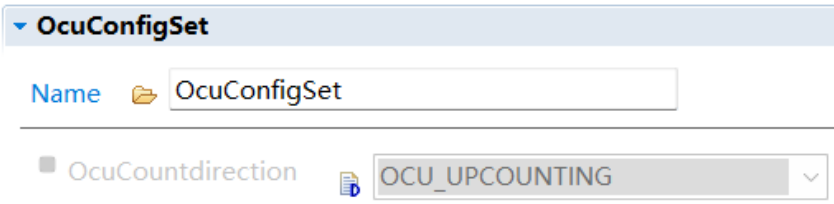
3.2.2.7 [OcuSetRelativeThresholdAp](#)

Variable	OcuSetRelativeThresholdAp	
Description	Adds/removes the service Ocu_SetRelativeThreshold() from the code.	
Screenshot	OcuSetRelativeThresholdApi    ▼	
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false

3.2.2.8 [OcuVersionInfoApi](#)

Variable	OcuVersionInfoApi	
Description	Switch to indicate that the Ocu_GetVersionInfo() is supported.	
Screenshot	OcuVersionInfoApi    ▼	
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false

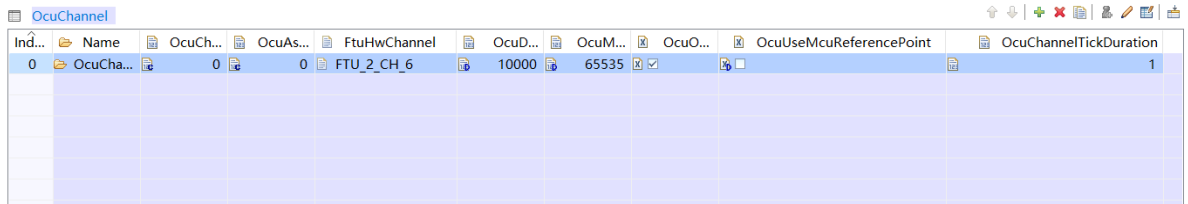
3.2.3 OcuConfigSet

Container	OcuConfigSet	
Description	This container is the base of a Configuration Set, which contains the configured OCU channels. This way, different configuration sets can be defined for post-build process.	
Screenshot		
Properties	Property	Value
	Type	Container

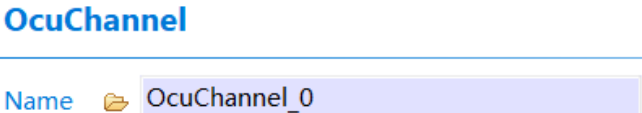
3.2.3.1 OcuCountdirection

Variable	OcuCountdirection	
Description	This parameter indicates the count direction for the whole OCU driver.	
Screenshot		
Properties	Property	Value
	Type	Variable: ENUMERATION
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false
	Range	OCU_DOWNCOUNTING, OCU_UPCOUNTING
	Default	OCU_UPCOUNTING

3.2.4 OcuChannel

Container	OcuChannel	
Description	N/A	
Screenshot		
Properties	Property	Value
	Type	MAP
	MIN	1

3.2.4.1 OcuChannel

Container	OcuChannel	
Description	Configuration of an individual OCU channel.	
Screenshot		

Properties	Property	Value
	Type	Container

3.2.4.1.1 OcuChannelId

Variable	OcuChannelId	
Description	Channel Id of the OCU channel. This value will be assigned to the symbolic name derived from the OcuChannel container short name. It defines the assignment of the channel to the physical OCU hardware channel.	
Screenshot		
Properties	Property	Value
	Type	Variable: INTEGER
	Range	0-65535

3.2.4.1.2 OcuAssignedHardwareChannel

Variable	OcuAssignedHardwareChannel	
Description	The physical hardware channel that is assigned to this logical channel.	
Screenshot		
Properties	Property	Value
	Type	Variable: INTEGER
	Range	0-255

3.2.4.1.3 FtuHwChannel

Variable	FtuHwChannel	
Description	The Ftu hardware channel that is assigned to this logical channel.	
Screenshot		
Properties	Property	Value
	Type	Variable: ENUMERATION
	Origin	FLAGCHIP

3.2.4.1.4 OcuDefaultThreshold

Variable	OcuDefaultThreshold	
Description	Value of comparison threshold used for Initialization.	
Screenshot		
Properties	Property	Value
	Type	Variable: INTEGER
	Origin	AUTOSAR_ECUC
	Range	0-16777215

3.2.4.1.5 OcuMaxCounterValue

Variable	OcuMaxCounterValue	
Description	Maximum value in ticks, the counter of the OCU channel is able to count.	

Screenshot		
Properties	Property	Value
	Type	Variable: INTEGER
	Origin	AUTOSAR_ECUC
	Range	1-16777215

3.2.4.1.6 OcuOuptutPinUsed

Variable	OcuOuptutPinUsed	
Description	Information about the usage of an output pin on this channel.	
Screenshot		
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	AUTOSAR_ECUC
	Default	false

3.2.4.1.7 OcuOutputPinDefaultState

Variable	OcuOutputPinDefaultState	
Description	The parameter OcuOutputPinDefaultState represents the state that a pin associated with a channel shall be set to after initialisation.	
Screenshot		
Properties	Property	Value
	Type	Variable: ENUMERATION
	Range	OCU_HIGH, OCU_LOW
	Default	OCU_LOW
	Origin	AUTOSAR_ECUC
	Optional	True

3.2.4.1.8 OcuOutputPinAction

Variable	OcuOutputPinAction	
Description	The parameter OcuOutputPinAction represents the action that a pin associated with a channel shall be set immediatly after a compare match event.	
Screenshot		
Properties	Property	Value
	Type	Variable: ENUMERATION
	Range	OCU_SET_LOW, OCU_SET_HIGH, OCU_TOGGLE, OCU_DISABLE
	Default	OCU_DISABLE
	Origin	FLAGCHIP
	Optional	True

3.2.4.1.9 OcuChannelTickDuration

Variable	OcuChannelTickDuration	
Description	Specifies the number of input clock edges (rising or falling edges) required to increase the channel counter by one (i.e. one counter tick). The value range depends on the used HW, not all values may be relevant	
Screenshot		
Properties	Property	Value
	Type	Variable: INTEGER
	Origin	AUTOSAR_ECUC

3.2.4.1.10 OcuNotification

Variable	OcuNotification	
Description	Definition of a function pointer to a Callback function.	
Screenshot		
Properties	Property	Value
	Type	FUNCTION-NAME
	Origin	AUTOSAR_ECUC
	Default	NULL_PTR

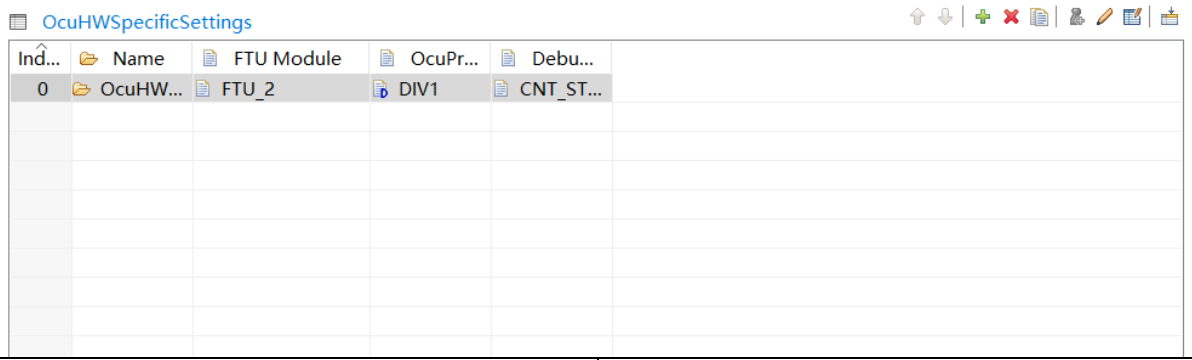
3.2.4.1.11 OcuHWSpecificSettingsRef

Variable	OcuHWSpecificSettingsRef	
Description	Reference to the OcuHWSpecificSettings used by the OcuChannel.	
Screenshot		
Properties	Property	Value
	Type	Reference to [OcuHWSpecificSettings]
	Origin	AUTOSAR_ECUC

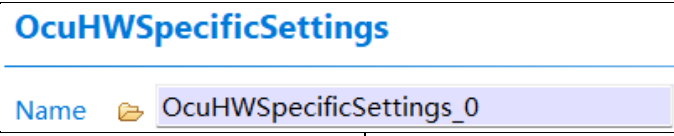
3.2.4.1.12 OcuChannelEcucPartitionRef

Variable	OcuChannelEcucPartitionRef	
Description	Maps an OCU channel to zero or multiple ECUC partitions to limit the access to this channel. The ECUC partitions referenced are a subset of the ECUC partitions where the OCU driver is mapped to.	
Screenshot		
Properties	Property	Value
	Type	Reference to [EcucPartition]
	Origin	AUTOSAR_ECUC

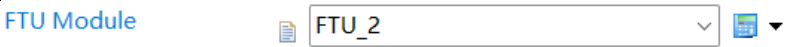
3.2.5 OcuHWSpecificSettings

Container	OcuHWSpecificSettings	
Description	N/A	
Screenshot		
Properties	Property	Value
	Type	MAP
	MIN	0

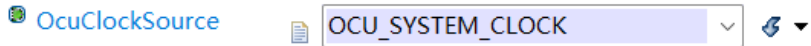
3.2.5.1 OcuHWSpecificSettings

Container	OcuHWSpecificSettings	
Description	This container contains Ocu-specific parameters for selecting the clock source and setting optional prescalers if supported by hardware.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.5.1.1 FTU Module

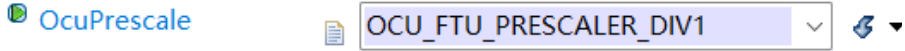
Variable	FTU Module	
Description	Selects one Ftu modules available on the platform.	
Screenshot		
Properties	Property	Value
	Type	Variable: ENUMERATION
	Origin	FLAGCHIP

3.2.5.1.2 OcuClockSource

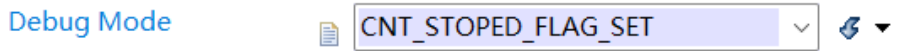
Variable	OcuClockSource	
Description	The OCU driver specific clock input for the unit can statically be configured to select different clock sources if provided by hardware. Enumeration literals are defined vendor specific.	
Screenshot		
Properties	Property	Value
	Type	Variable: ENUMERATION
	Range	OCU_SYSTEM_CLOCK, OCU_EXTERNAL_CLOCK

	Origin	FLAGCHIP
--	--------	----------

3.2.5.1.3 OcuPrescale

Variable	OcuPrescale	
Description	Optional OCU driver specific clock prescale factor, if supported by hardware. Implementation is defined vendor specific.	
Screenshot		
Properties	Property	Value
	Type	Variable: ENUMERATION
	Default	OCU_FTU_PRESCALER_DIV1
	Origin	FLAGCHIP

3.2.5.1.4 Debug Mode

Variable	Debug Mode	
Description	These bits allow select the behavior of the FTU counter, the CH(n)F bit, the channels output, and the writing to the MOD, CNTIN, and C(n)V registers (wave form register).	
Screenshot		
Properties	Property	Value
	Type	Variable: ENUMERATION
	Range	CNT_ON_OUTPUTS_ON, CNT_STOPED_OUTPUTS_FROZEN, CNT_STOPED_OUTPUTS_SAFE, CNT_STOPED_FLAG_SET
	Origin	FLAGCHIP

3.2.6 OcuGeneral

Container	OcuGeneral	
Description	This container contains the module-wide configuration parameters of the OCU Driver.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.6.1 OcuDevErrorDetect

Variable	OcuDevErrorDetect	
Description	Switches the development error detection and notification on or off.	
Screenshot		
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	AUTOSAR_ECUC

	SymbolicNameValue	false
--	-------------------	-------

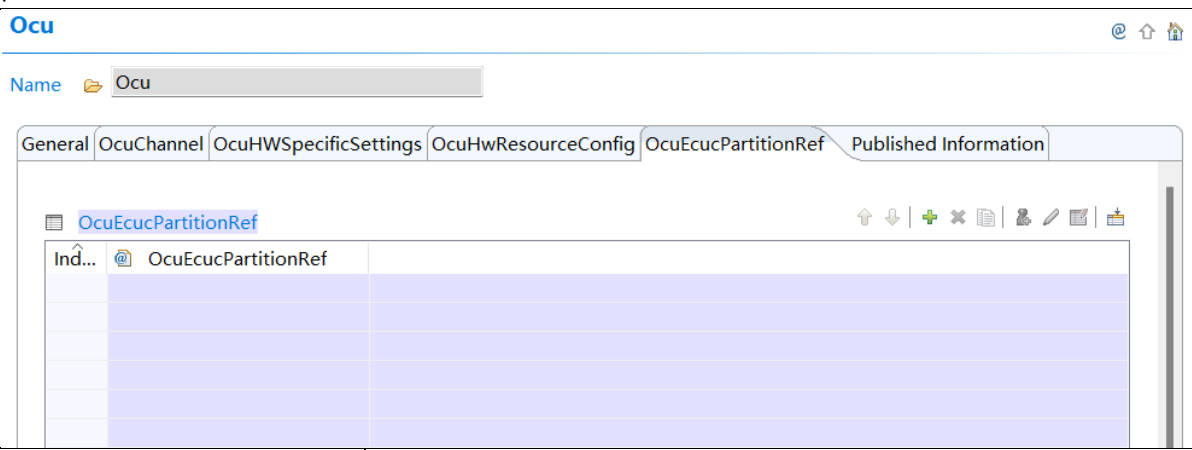
3.2.6.2 OcuDisableRuntimeErrorDetect

Variable	OcuDisableRuntimeErrorDetect	
Description	Switches the runtime error detection and notification on or off.	
Screenshot		
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false

3.2.6.3 Ocu Multicore Support

Variable	OcuMulticoreSupport	
Description	Switches the multicore support on or off.	
Screenshot		
Properties	Property	Value
	Type	Variable: BOOLEAN
	Origin	FLAGCHIP
	SymbolicNameValue	false

3.2.6.4 OcuEcucPartitionRef

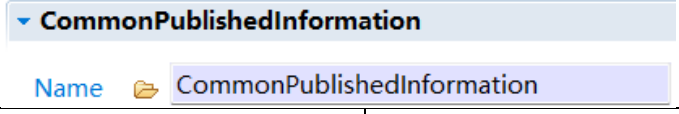
Variable	OcuEcucPartitionRef	
Description	Maps the OCU driver to zero or multiple ECUC partitions to make the driver API available in the according partition.	
Screenshot		
Properties	Property	Value
	Type	Reference to [EcucPartition]
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false

3.2.6.5 OcuKernelEcucPartitionRef

Variable	OcuKernelEcucPartitionRef	
Description	Maps the OCU kernel to zero or one ECUC partitions to assign the driver kernel to a certain core. The ECUC	

	partition referenced is a subset of the ECUC partitions where the OCU driver is mapped to.	
Screenshot		
Properties	Property	Value
	Type	Reference to [EcucPartition]
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false

3.2.7 CommonPublishedInformation

Container	CommonPublishedInformation	
Description	Common container, aggregated by all modules. It contains published information about vendor and versions.	
Screenshot		
Properties	Property	Value
	Type	Container

3.2.7.1 ArReleaseMajorVersion

Variable	ArReleaseMajorVersion	
Description	Major version number of AUTOSAR specification on which the appropriate implementation is based on.	
Screenshot		
Properties	Property	Value
	Type	Variable: INTEGER_LABEL
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Default	4

3.2.7.2 ArReleaseMinorVersion

Variable	ArReleaseMinorVersion	
Description	Minor version number of AUTOSAR specification on which the appropriate implementation is based on.	
Screenshot		
Properties	Property	Value
	Type	Variable: INTEGER_LABEL
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Default	6

3.2.7.3 ArReleaseRevisionVersion

Variable	ArReleaseRevisionVersion	
Description	Revision version number of AUTOSAR specification on which the appropriate implementation is based.	
Screenshot		
Properties	Property	Value

	Type	Variable: INTEGER_LABEL
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Default	0

3.2.7.4 SwMajorVersion

Variable	SwMajorVersion	
Description	Major version number of the vendor specific implementation of the module. The numbering is vendor specific.	
Screenshot		
Properties	Property	Value
	Type	Variable: INTEGER_LABEL
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Default	0

3.2.7.5 SwMinorVersion

Variable	SwMinorVersion	
Description	Minor version number of the vendor specific implementation of the module. The numbering is vendor specific.	
Screenshot		
Properties	Property	Value
	Type	Variable: INTEGER_LABEL
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Default	6

3.2.7.6 SwPatchVersion

Variable	SwPatchVersion	
Description	Patch level version number of the vendor specific implementation of the module. The numbering is vendor specific.	
Screenshot		
Properties	Property	Value
	Type	Variable: INTEGER_LABEL
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Default	0

3.2.7.7 ModuleId

Variable	ModuleId	
Description	Module ID of this module from Module List.	

Screenshot		
Properties	Property	Value
	Type	Variable: INTEGER_LABEL
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Default	125

3.2.7.8 VendorId

Variable	VendorId	
Description	Vendor ID of the dedicated implementation of this module according to the AUTOSAR vendor list.	
Screenshot		
Properties	Property	Value
	Type	Variable: INTEGER_LABEL
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Default	174

Chapter 4 Configuration Guides

4.1 Configuration Item Constraint

1. If the OCU module uses the same FTU instance with PWM/ICU/GPT module, the configuration tool will report an error.

GptChannelConfigSet

Name

GptChannelConfigSet

GptChannelConfiguration

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Ind...	Name	GptCh...	GptHwC...	GptCh...	GptCh...	GptCh...	GptCh...	GptCh...	GptCh...	GptDb...	Gptt...
0	GptCha...	0	FTU_4_CH...	GPT_C...	2.4E7	/Gpt/G...	65535	☒	☐	☐	☐
1	GptCha...	1	FTU_1_CH...	GPT_C...	750000.0	/Gpt/G...	65535	☐	☐	☐	☐

OcuHWSpecificSettings

@  

Name

 OcuHWSpecificSettings_2

General

FTU Module

 FTU_4  

OcuClockSource

 OCU_FTU_BUS_CLK  

OcuPrescale

 OCU_FTU_PRESCALER_DIV1  

Debug Mode

 CNT_ON_OUTPUTS_ON  

@ ↑ 🏠

2. If the number `OcuEcucPartitionRef` = 0, the check-box `OcuMulticoreEnabled` must be 'false'.

[illegible]

Name Ocu

General OcuChannel OcuHWSpecificSettings OcuHwResourceConfig OcuEcucPartitionRef Published Information

OcuSetRelativeThresholdApi

OcuVersionInfoApi

▼ OcuConfigSet

Name OcuConfigSet

OcuCountdirection

OCU_UPCOUNTING

▼ OcuGeneral

Name OcuGeneral

OcuDevErrorDetect

Ocu Multicore Support

OcuKernelEcucPartitionRef

3. Different channels of the same instance must be assigned to the same partition.

Name Ocu

General OcuChannel OcuHWSpecificSettings OcuHwResourceConfig OcuEcucPartitionRef Published Information

OcuChannel

Ind...	Name	OcuCh...	OcuAs...	OcuHWSpecificSettingsRef	OcuHwChannel	OcuD...	OcuM...	OcuO...	OcuC...
0	OcuCha...	0	0	/Ocu/Ocu/OcuConfigSet/OcuHWSpecificSettings_0	FTU_0_CH_0	10000	60000		
1	OcuCha...	1	1	/Ocu/Ocu/OcuConfigSet/OcuHWSpecificSettings_1	FTU_3_CH_3	10000	40000		
2	OcuCha...	2	2	/Ocu/Ocu/OcuConfigSet/OcuHWSpecificSettings_0	FTU_0_CH_5	10000	60000		

OcuChannel

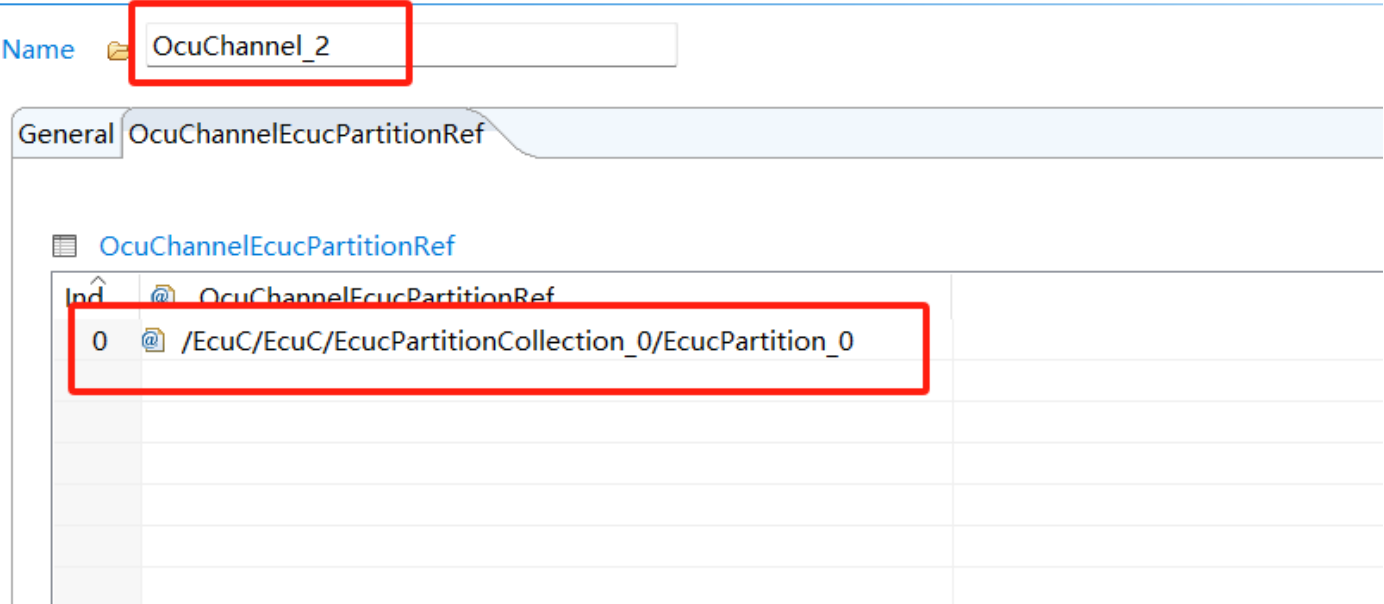
Name OcuChannel_0

General OcuChannelEcucPartitionRef

OcuChannelEcucPartitionRef

Ind	OcuChannelEcucPartitionRef
0	/EcuC/EcuC/EcucPartitionCollection_0/EcucPartition_0

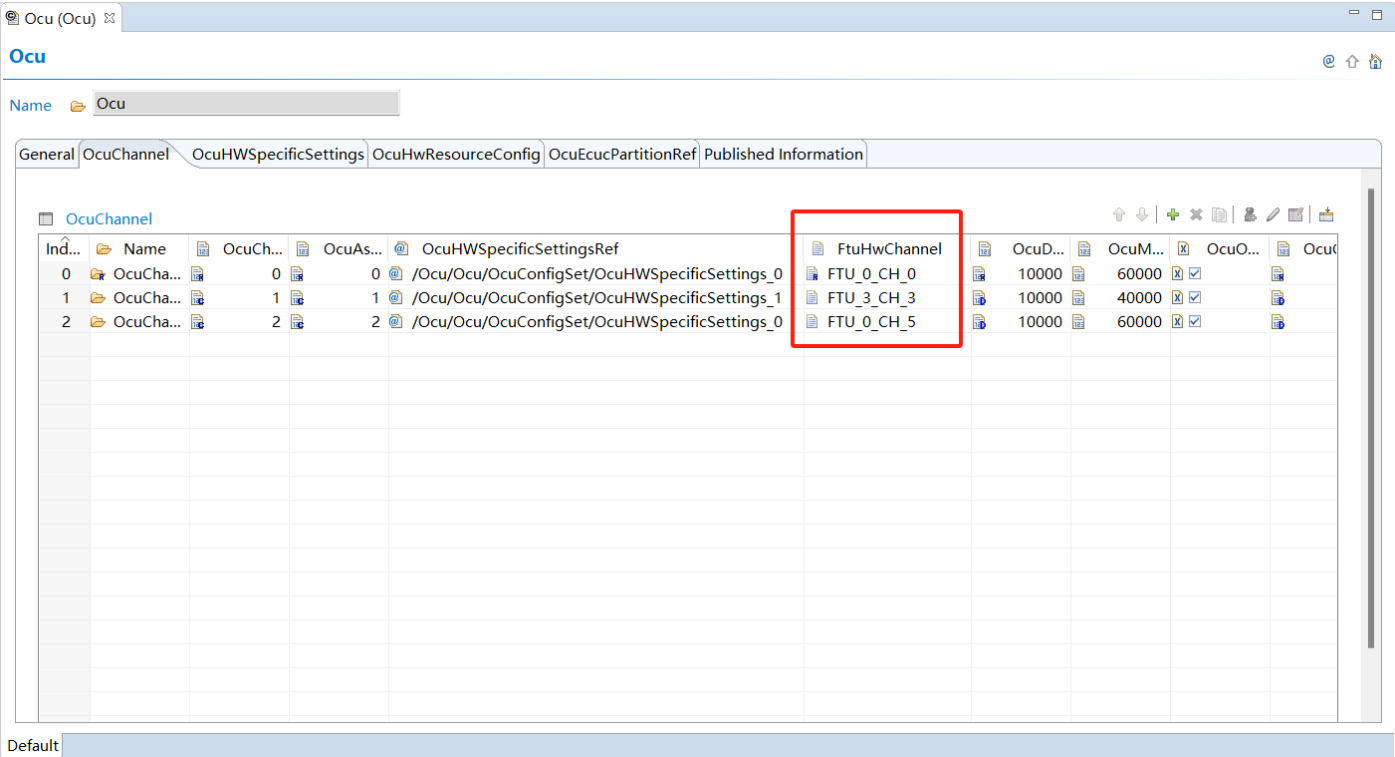
OcuChannel



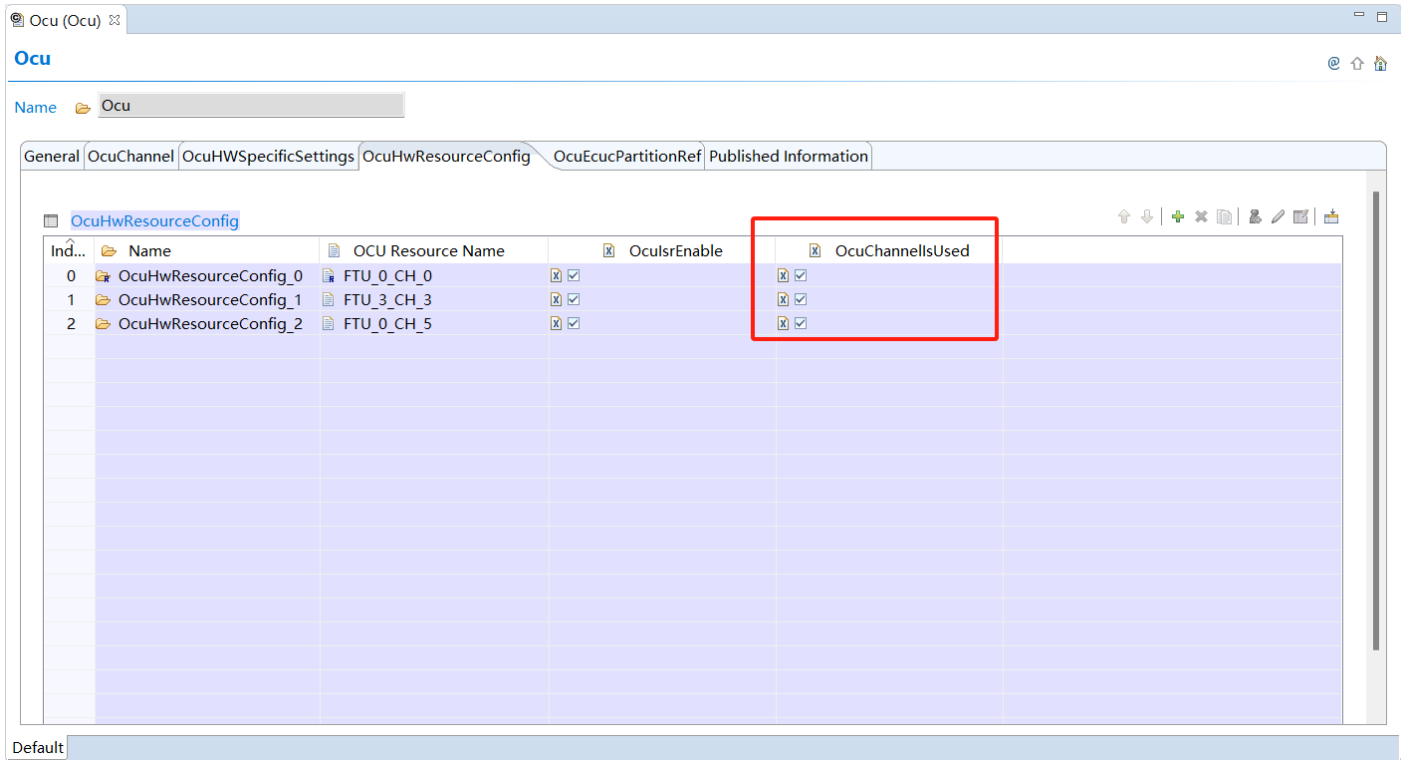
4.2 OCU Usage Common Steps

Basically, the OCU module can be configured by the following 4 steps:

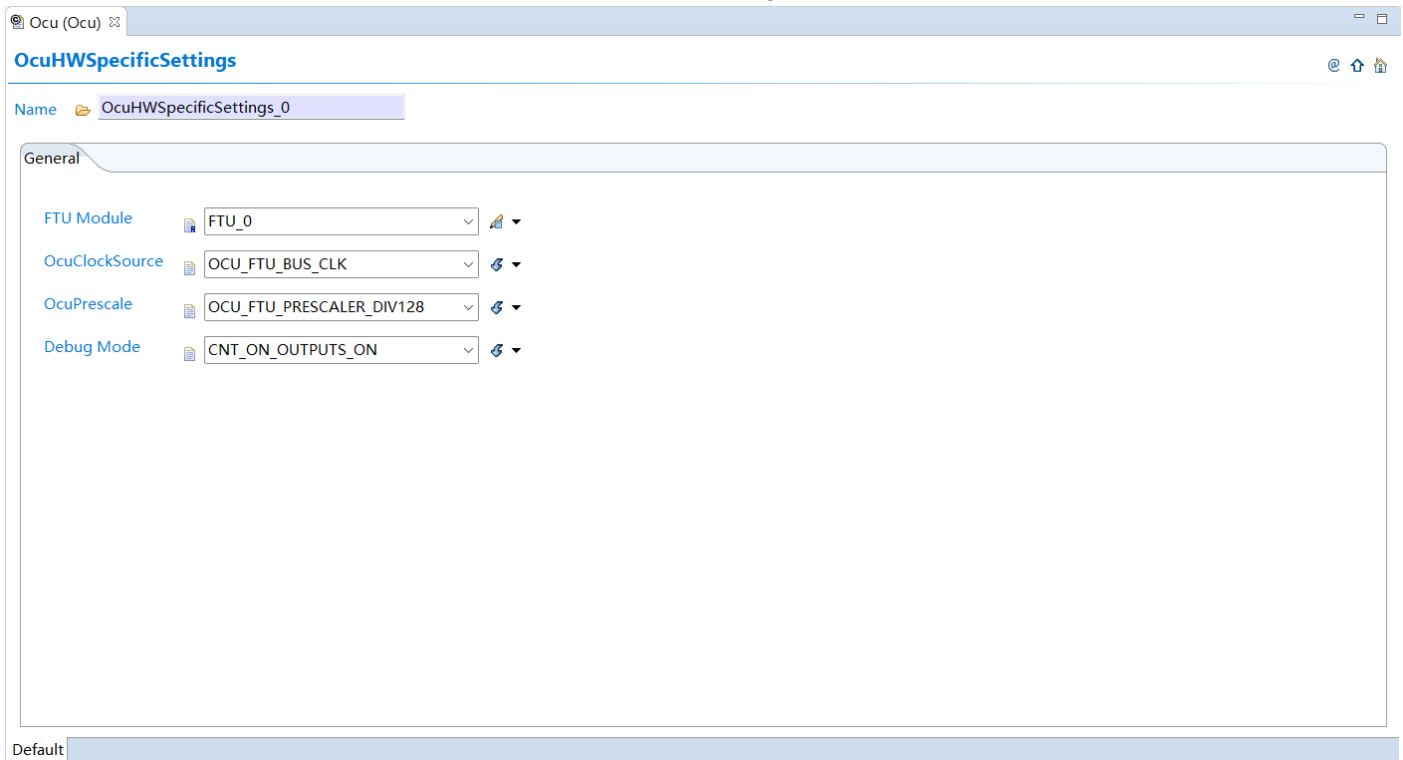
- Add a module object in the OcuHWSpecificSettings tab and a channel object in the OcuChannel tab. The added channel needs to be included in the module.



- Enable the selected channel in OcuHwResourceConfig tab.



- Configure OCU module parameters in the OcuHWSpecificSettings tab.



- Configure OCU channel parameters in the OcuChannel tab.

Ocu (Ocu)

OcuChannel

Name OcuChannel_0

General OcuChannelEcucPartitionRef

OcuChannelId (0 -> 65535)	0
OcuAssignedHardwareChannel (0 -> 255)	0
OcuHWSpecificSettingsRef	/Ocu/Ocu/OcuConfigSet/OcuHWSpecificSettings_0
FtuHwChannel	FTU_0_CH_0
OcuDefaultThreshold (0 -> 65535)	10000
OcuHardwareTriggeredAdc (0 -> 255)	0
OcuHardwareTriggeredDMA (0 -> 255)	0
OcuMaxCounterValue (1 -> 65535)	60000
OcuNotification	Ocu_Notification_channel_0
OcuOutputPinUsed	☒
OcuOutputPinDefaultState	OCU_LOW
OcuOutputPinAction	OCU_TOGGLE
OcuChannelTickDuration (1 -> 32768)	1

Default

- Choose which API will be used.

Ocu (Ocu)

Ocu

Name Ocu

General OcuChannel OcuHWSpecificSettings OcuHwResourceConfig OcuEcucPartitionRef Published Information

Config Variant VariantPostBuild

OcuConfigurationOfOptionalApis

Name OcuConfigurationOfOptionalApis	
OcuDeInitApi ☒	OcuGetCounterApi ☒
OcuNotificationSupported ☒	OcuSetAbsoluteThresholdApi ☒
OcuSetPinActionApi ☒	OcuSetPinStateApi ☒
OcuSetRelativeThresholdApi ☒	OcuVersionInfoApi ☒

OcuConfigSet

Name OcuConfigSet

OcuCountdirection OCU_UPCOUNTING

OcuGeneral

Name OcuGeneral

Default

- Enable multicore and allocate EcucPartitions for each channel (Skip this step if multicore is disable)

The image displays two screenshots of the Ocu configuration tool interface.

Top Screenshot: General Tab

The **General** tab is selected. The **Name** field is set to **Ocu**. Below the tabs, the **OcuConfigurationOfOptionalApis** section lists several APIs with checkboxes and edit icons:

- OcuDelnitApi** (checked)
- OcuNotificationSupported** (checked)
- OcuSetPinActionApi** (checked)
- OcuSetRelativeThresholdApi** (checked)
- OcuGetCounterApi** (checked)
- OcuSetAbsoluteThresholdApi** (checked)
- OcuSetPinStateApi** (checked)
- OcuVersionInfoApi** (checked)

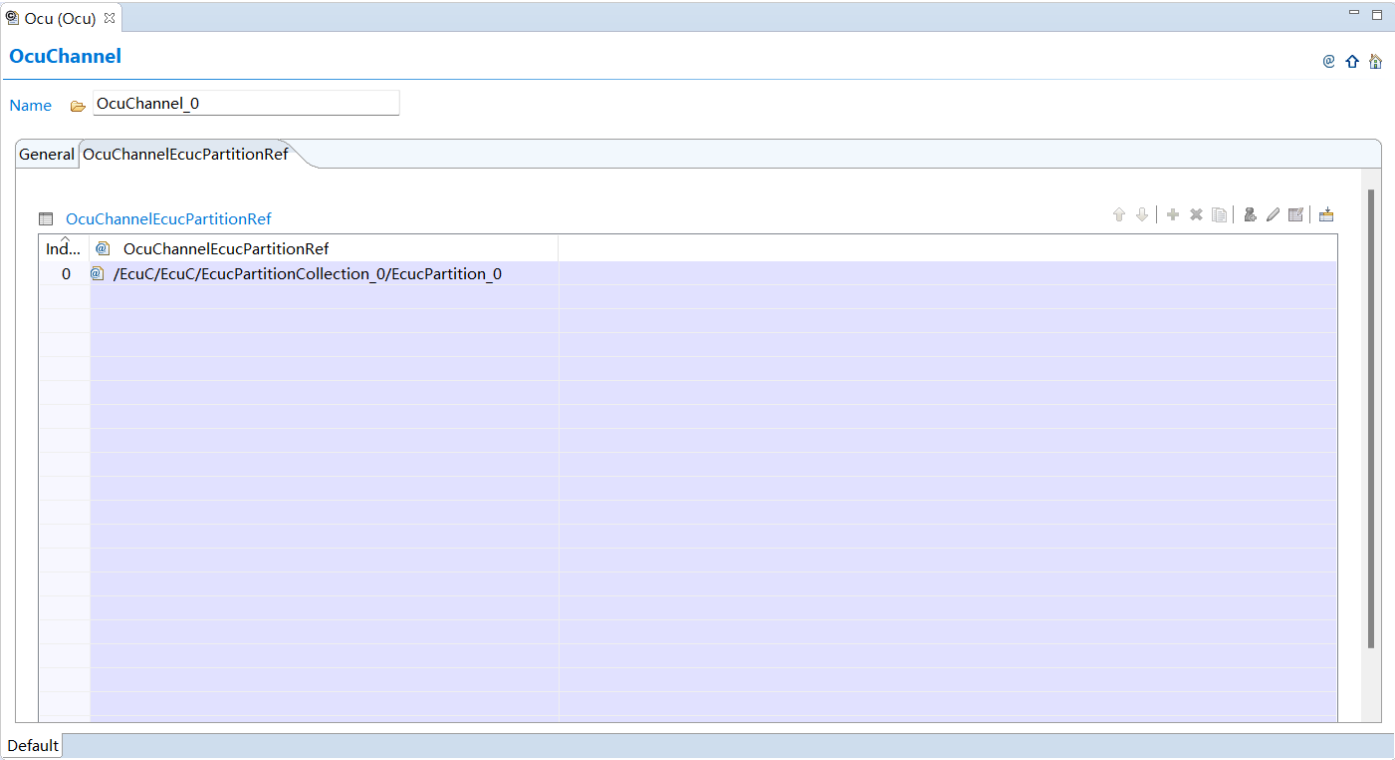
The **OcuConfigSet** section shows the **Name** field set to **OcuConfigSet** and the **OcuCountdirection** dropdown set to **OCU_UPCOUNTING**.

The **OcuGeneral** section shows the **Name** field set to **OcuGeneral**. The **OcuDevErrorDetect** checkbox is checked. The **OcuMulticoreSupport** checkbox is checked and highlighted with a red box. The **OcuKernelEcucPartitionRef** dropdown is set to **OcuKernelEcucPartitionRef**.

Bottom Screenshot: OcuEcucPartitionRef Table

The **OcuEcucPartitionRef** table is displayed. The table has two columns: **Ind...** and **OcuEcucPartitionRef**. The table contains two rows:

Ind...	OcuEcucPartitionRef
0	/EcuC/EcuC/EcucPartitionCollection_0/EcucPartition_0
1	/EcuC/EcuC/EcucPartitionCollection_0/EcucPartition_1



- Generate configuration files.